

Competitive Programming Notebook

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1 Miscellaneous

1.1 Template

```
1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 //define int long long
6 #define ll long long
7 #define ld long double
8 #define fi first
9 #define se second
10 #define pb push_back
11 #define eb emplace_back
12 #define all(x) x.begin(), x.end()
13 #define rall(x) x.rbegin(), x.rend()
14 #define ones(n) __builtin_popcountll(n)
15 #define msb(n) (63 - __builtin_clzll(n))
16 #define lsb(n) __builtin_ctzll(n)
17
18 const int N = 2e5 + 5, M = 1e3 + 5, LOG = 31;
19 const int inf = 0x3f3f3f3f;
20 const ll llinf = 0x3f3f3f3f3f3f3f3f;
21 const int MOD = 1e9 + 7;
22 const double eps = 1e-9, PI = acos(-1);
23
24 void testCase() {
25 }
26
27 void preCompute() {
28 }
29
30 int32_t main() {
31 #ifdef C_Lion
32     freopen("input.txt", "r", stdin);
33     freopen("output.txt", "w", stdout);
34     freopen("errors.txt", "w", stderr);
35 #else
36     //     freopen("input.in", "r", stdin);
37     //     freopen("output.out", "w", stdout);
38 #endif
39
40     ios_base::sync_with_stdio(false), cin.tie(nullptr), cout.tie(
        nullptr);
41     cout << fixed << setprecision(static_cast<int32_t>(-log10(eps)));
42     preCompute();
43
44     int tc = 1;
45     cin >> tc;
46     for (int TC = 1; TC <= tc; TC++) {
47         // cout << "Case #" << TC << ": ";
48         testCase();
49     }
50 }
51
52 /*
53
54
55
56
57
58 */
```

1.2 fastIO

```
1 const int MOD = 1e9 + 7;
2 const int BUF_SZ = 1 << 15;
3
4 inline namespace Input {
5     char buf[BUF_SZ];
6     int pos;
7     int len;
8     char next_char() {
9         if (pos == len) {
10             pos = 0;
11             len = (int)fread(buf, 1, BUF_SZ, stdin);
12             if (!len) { return EOF; }
13         }
14         return buf[pos++];
15     }
16
17     int read_int() {
18         int x;
19         char ch;
20         int sgn = 1;
21         while (!isdigit(ch = next_char())) {
22             if (ch == '-') { sgn *= -1; }
23         }
24         x = ch - '0';
25         while (isdigit(ch = next_char())) { x = x * 10 + (ch - '0'); }
26         return x * sgn;
27     }
28 } // namespace Input
29 inline namespace Output {
30     char buf[BUF_SZ];
31     int pos;
32
33     void flush_out() {
```

```
34         fwrite(buf, 1, pos, stdout);
35         pos = 0;
36     }
37
38     void write_char(char c) {
39         if (pos == BUF_SZ) { flush_out(); }
40         buf[pos++] = c;
41     }
42
43     void write_int(int x) {
44         static char num_buf[100];
45         if (x < 0) {
46             write_char('-');
47             x *= -1;
48         }
49         int len = 0;
50         for (; x >= 10; x /= 10) { num_buf[len++] = (char)('0' + (x %
            10)); }
51         write_char((char)('0' + x));
52         while (len) { write_char(num_buf[--len]); }
53         write_char('\n');
54     }
55
56     // auto-flush output when program exits
57     void init_output() { assert(atexit(flush_out) == 0); }
58 } // namespace Output
```

1.3 grid

```
1 int dx[] = {0, 0, -1, 1, -1, -1, 1, 1};
2 int dy[] = {-1, 1, 0, 0, -1, 1, -1, 1};
3 char di[] = {'L', 'R', 'U', 'D'};
```

1.4 random

```
1 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2 template <typename T>
3 T random(T l, T r) {
4     return uniform_int_distribution(l, r)(rng);
5 }
6
7 vector<int> random(int n) {
8     vector<int> v(n);
9     // shuffle 1
10    shuffle(v.begin(), v.end(), rng);
11    // shuffle 2
12    for (int i = 1; i < n; i++)
13        swap(v[i], v[uniform_int_distribution<int>(0, i)(rng)]);
14    return v;
15 }
```

1.5 coordinateCompression

```
1 struct Compress {
2     map<int, int> m;
3     vector<int> undo;
4
5     Compress(vector<int>& v, int base = 0) {
6         m.clear();
7         set<int> s;
8         for (int x : v) {
9             s.insert(x);
10        }
11        undo.resize(s.size() + base);
12        int i = base;
13        for (int x : s) {
14            undo[i] = x;
15            m[x] = i++;
16        }
17        for (int& x : v)
18            x = m[x];
19    }
20 };
```

1.6 grayCode

```
1 vector<int> grayCode(int n) {
2     vector<int> vec = {0};
3     vector<bool> used(1 << n, false);
4     used[0] = true;
5     while (vec.size() < (1 << n)) {
6         int bit = 0;
7         while (used[vec.back() ^ (1 << bit)])
8             bit++;
9         vec.push_back(vec.back() ^ (1 << bit));
10        used[vec.back() ^ (1 << bit)] = true;
11    }
12    return vec;
13 }
```

1.7 customHashHashTable

```
1 #include <ext/pb_ds/assoc_container.hpp>
2 using namespace __gnu_pbds;
3 //gp_hash_table<int, int> table;
4 const int RANDOM = chrono::high_resolution_clock::now().
    time_since_epoch().count();
5 struct chash {
6     int operator()(int x) const { return x ^ RANDOM; }
7 };
8 // gp_hash_table<int, int, chash> table;
```

1.8 pragmas

```
1 // GCC Optimizations
2 #pragma GCC optimize("O3,unroll-loops")
3 #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

2 Data Structures (DS)

2.1 Segment Tree

2.1.1 SegmentTree

```
1 struct SEG {
2     ll sum = 0;
3     SEG() {
4     }
5     SEG(ll x){
6         sum = x;
7     }
8 };
9 struct segTree {
10     vector<SEG> seg;
11     int sz = 1,n;
12     segTree(int nn){
13         n = nn;
14         while (sz < nn)
15             sz *= 2;
16         seg.assign(2 * sz , SEG());
17     }
18
19     SEG merge(SEG seg1, SEG seg2) {
20         SEG ret;
21         ret.sum = seg1.sum + seg2.sum;
22         return ret;
23     }
24
25     void build(vector<int> &v,int x, int lx, int rx) {
26         if (lx == rx) {
27             seg[x] = SEG(v[lx]);
28             return;
29         }
30         int mid = (lx + rx) / 2;
31         int lf = 2 * x + 1, rt = 2 * x + 2;
32
33         build(v,lf, lx, mid);
34         build(v,rt, mid + 1, rx);
35         seg[x] = merge(seg[lf], seg[rt]);
36     }
37     void build(vector<int> &v){
38         build(v,0, 0, n-1);
39     }
40
41     void update(int i, ll val, int x, int lx, int rx) {
42         if (lx == rx) {
43             seg[x] = SEG(val);
44             return;
45         }
46         int mid = (lx + rx) / 2,lf = 2*x+1,rt= 2*x+2;
47         if(i <= mid)
48             update(i, val, lf, lx, mid);
49         else
50             update(i, val, rt, mid + 1, rx);
51         seg[x] = merge(seg[lf] , seg[rt]);
52     }
53
54     void update(int i, ll val) {
55         update(i, val, 0, 0, n-1);
56     }
57
58     SEG query(int l, int r, int x, int lx, int rx) {
59         if (l <= lx && r >= rx)
60             return seg[x];
61         if (l > rx || r < lx)
62             return SEG();
63
64         int mid = (lx + rx) / 2,lf = 2 * x + 1,rt = 2*x+2;
65
66         return merge(query(l, r, lf, lx, mid) , query(l, r, rt, mid +
67             1, rx));
68     }
69     SEG query(int l, int r) {
70         return query(l, r, 0, 0, n-1);
71     }
72 };
```

2.1.2 LazySegmentTree

```
1 struct SEG {
2     ll sum = 0, lz = 0;
3
4     SEG() {
5     }
6
7     SEG(ll x) {
8         sum = x;
9     }
10 };
11
12 struct segTree {
13     vector<SEG> seg;
14     int sz = 1, n;
15     ll NO_OP = 0;
16
17     segTree(int nn) {
18         n = nn;
19         while (sz < nn)
20             sz *= 2;
21         seg.assign(2 * sz, SEG());
22     }
23
24     SEG merge(SEG seg1, SEG seg2) {
25         SEG ret;
26         ret.sum = seg1.sum + seg2.sum;
27         return ret;
28     }
29
30     void build(vector<int> &v, int x, int lx, int rx) {
31         if (lx == rx) {
32             seg[x] = SEG(v[lx]);
```

```
33         return;
34     }
35     int mid = (lx + rx) / 2;
36     int lf = 2 * x + 1, rt = 2 * x + 2;
37
38     build(v, lf, lx, mid);
39     build(v, rt, mid + 1, rx);
40     seg[x] = merge(seg[lf], seg[rt]);
41 }
42
43 void build(vector<int> &v) {
44     build(v, 0, 0, n - 1);
45 }
46
47 void push(int x, int lx, int rx) {
48     if (seg[x].lz != NO_OP) {
49         seg[x].sum += seg[x].lz * (rx - lx + 1);
50         int lf = 2 * x + 1, rt = 2 * x + 2;
51         if (lx != rx) {
52             seg[lf].lz += seg[x].lz;
53             seg[rt].lz += seg[x].lz;
54         }
55         seg[x].lz = NO_OP;
56     }
57 }
58
59 void update(int l, int r, ll val, int x, int lx, int rx) {
60     push(x, lx, rx);
61     if (l > rx || r < lx)
62         return;
63     if (l <= lx && r >= rx) {
64         seg[x].lz += val;
65         push(x, lx, rx);
66         return;
67     }
68     int mid = (lx + rx) / 2, lf = 2 * x + 1, rt = 2 * x + 2;
69     update(l, r, val, lf, lx, mid);
70     update(l, r, val, rt, mid + 1, rx);
71     seg[x] = merge(seg[lf], seg[rt]);
72 }
73
74 void update(int l, int r, ll val) {
75     update(l, r, val, 0, 0, sz - 1);
76 }
77
78 SEG query(int l, int r, int x, int lx, int rx) {
79     push(x, lx, rx);
80     if (l <= lx && r >= rx)
81         return seg[x];
82     if (l > rx || r < lx)
83         return SEG();
84
85     int mid = (lx + rx) / 2, lf = 2 * x + 1, rt = 2 * x + 2;
86
87     return merge(query(l, r, lf, lx, mid), query(l, r, rt, mid + 1,
88         rx));
89 }
90
91 SEG query(int l, int r) {
92     return query(l, r, 0, 0, sz - 1);
93 }
```

2.1.3 PersistentSegmentTree

```
1 struct Node {
2     Node *l, *r;
3     ll val = 0;
4
5     Node(int val) : l(NULL), r(NULL), val(val) {
6     }
7
8     Node() : l(NULL), r(NULL) {
9     }
10
11     Node(Node* l, Node* r) : l(l), r(r) {
12         if (l != NULL) val += l->val;
13         if (r != NULL) val += r->val;
14     }
15
16     void addChild() {
17         l = new Node();
18         r = new Node();
19     }
20 };
21
22 struct PST {
23     int n;
24
25     PST(int n) : n(n + 1) {
26     }
27
28     Node merge(Node x, Node y) {
29         Node ret;
30         ret.val = x.val + y.val;
31         return ret;
32     }
33
34     Node* update(Node* v, int i, int val, int lx, int rx) {
35         if (lx == rx) return new Node(v->val + val);
36         int mid = (lx + rx) / 2;
37         if (!v->l) v->addChild();
38         if (i <= mid) return new Node(update(v->l, i, val, lx, mid), v
39             ->r);
40         return new Node(v->l, update(v->r, i, val, mid + 1, rx));
41     }
42
43     Node* update(Node* v, int i, int val) { return update(v, i, val, 0,
44         n - 1); }
45
46     Node query(Node* v, int l, int r, int lx, int rx) {
47         if (l > rx || r < lx) return {};
```

```

48     int mid = (lx + rx) / 2;
49     return merge(query(v->l, l, r, lx, mid), query(v->r, l, r, mid
50         + 1, rx));
51 }
52 Node query(Node* v, int l, int r) { return query(v, l, r, 0, n - 1)
53     ; }
54 int getKth(Node* a, Node* b, int k, int lx, int rx) {
55     if (lx == rx) return lx;
56     if (!a->l) a->addChild();
57     if (!b->l) b->addChild();
58     int rem = b->l->val - a->l->val;
59     int mid = (lx + rx) / 2;
60     if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
61     return getKth(a->r, b->r, k - rem, mid + 1, rx);
62 }
63
64 int getKth(Node* a, Node* b, int k) { return getKth(a, b, k, 0, n -
65     1); }

```

2.1.4 PersistentSegTreeVector

```

1 struct Node {
2     Node *l, *r;
3     int val = 0;
4
5     Node(int val): l(NULL), r(NULL), val(val) {}
6     Node(): l(NULL), r(NULL) {}
7     Node(Node *l, Node *r): l(l), r(r) {
8         if (l != NULL) val += l->val;
9         if (r != NULL) val += r->val;
10    }
11
12    void addChild() {
13        l = new Node();
14        r = new Node();
15    }
16 };
17
18 struct PST {
19     int n;
20     PST(int n): n(n + 1) {}
21
22     Node merge(Node x, Node y) {
23         Node ret;
24         ret.val = x.val + y.val;
25         return ret;
26     }
27
28     Node *set(Node *v, int i, int val, int lx, int rx) {
29         if (lx == rx) return new Node(val);
30         int mid = (lx + rx) / 2;
31         if (!v->l) v->addChild();
32         if (i <= mid) return new Node(set(v->l, i, val, lx, mid), v->r)
33             ;
34         return new Node(v->l, set(v->r, i, val, mid + 1, rx));
35     }
36
37     Node *set(Node *v, int i, int val) { return set(v, i, val, 0, n -
38         1); }
39
40     // [l, r] r is included
41     Node query(Node *v, int l, int r, int lx, int rx) {
42         if (l > rx || r < lx) return {};
43         if (l <= lx && r >= rx) return *v;
44         if (!v->l) v->addChild();
45         int mid = (lx + rx) / 2;
46         return merge(query(v->l, l, r, lx, mid), query(v->r, l, r, mid
47             + 1, rx));
48     }
49
50     Node query(Node *v, int l, int r) { return query(v, l, r, 0, n - 1)
51         ; }
52
53     int getKth(Node *a, Node *b, int k, int lx, int rx) {
54         if (lx == rx) return lx;
55         if (!a->l) a->addChild();
56         if (!b->l) b->addChild();
57         int rem = b->l->val - a->l->val;
58         int mid = (lx + rx) / 2;
59         if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
60         return getKth(a->r, b->r, k - rem, mid + 1, rx);
61     }
62
63     int getKth(Node *a, Node *b, int k) { return getKth(a, b, k, 0, n -
64         1); }
65 };

```

2.1.5 SegmentTreeBeats

```

1 struct Node {
2     int mn;
3     int mnCnt;
4     int secondMn;
5
6     int mx;
7     int mxCnt;
8     int secondMx;
9
10    int sum;
11
12    int lazyAdd;
13    int lazyEqual;
14    bool isLazyEqual;
15
16    Node() {
17        mx = -inf;
18        secondMx = -inf;
19        mxCnt = 0;
20
21        mn = inf;
22        secondMn = inf;

```

```

23        mnCnt = 0;
24
25        sum = 0;
26
27        lazyAdd = 0;
28        lazyEqual = 0;
29        isLazyEqual = 0;
30    };
31
32    Node(int x) {
33        mx = x;
34        secondMx = -inf;
35        mxCnt = 1;
36
37        mn = x;
38        secondMn = inf;
39        mnCnt = 1;
40
41        sum = x;
42
43        lazyAdd = 0;
44        lazyEqual = 0;
45        isLazyEqual = 0;
46    }
47 };
48
49 struct SegTree {
50     int tree_size;
51     vector<Node> seg_data;
52
53     SegTree(int n) {
54         tree_size = 1;
55         while (tree_size < n) tree_size *= 2;
56         seg_data.resize(2 * tree_size, Node());
57     }
58
59     Node merge(Node& lf, Node& ri) {
60         Node ans = Node();
61
62         ans.sum = lf.sum + ri.sum;
63
64         ans.mx = max(lf.mx, ri.mx);
65         ans.secondMx = max(lf.secondMx, ri.secondMx);
66         ans.mxCnt = 0;
67         if (lf.mx == ans.mx) ans.mxCnt += lf.mxCnt;
68         else ans.secondMx = max(ans.secondMx, lf.mx);
69         if (ri.mx == ans.mx) ans.mxCnt += ri.mxCnt;
70         else ans.secondMx = max(ans.secondMx, ri.mx);
71
72         ans.mn = min(lf.mn, ri.mn);
73         ans.secondMn = min(lf.secondMn, ri.secondMn);
74         ans.mnCnt = 0;
75         if (lf.mn == ans.mn) ans.mnCnt += lf.mnCnt;
76         else ans.secondMn = min(ans.secondMn, lf.mn);
77         if (ri.mn == ans.mn) ans.mnCnt += ri.mnCnt;
78         else ans.secondMn = min(ans.secondMn, ri.mn);
79
80         return ans;
81     }
82
83     void init(vector<int>& nums, int ni, int lx, int rx) {
84         if (rx - lx == 1) {
85             if (lx < nums.size())
86                 seg_data[ni] = Node(nums[lx]);
87             return;
88         }
89
90         int mid = lx + (rx - lx) / 2;
91         init(nums, 2 * ni + 1, lx, mid);
92         init(nums, 2 * ni + 2, mid, rx);
93
94         seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
95             2]);
96     }
97
98     void init(vector<int>& nums) {
99         init(nums, 0, 0, tree_size);
100     }
101
102     void doPushEq(int ni, int lx, int rx, int x) {
103         seg_data[ni].mn = seg_data[ni].mx = seg_data[ni].lazyEqual = x;
104         seg_data[ni].isLazyEqual = 1;
105         seg_data[ni].mnCnt = seg_data[ni].mxCnt = rx - lx;
106         seg_data[ni].secondMx = -inf;
107         seg_data[ni].secondMn = inf;
108
109         seg_data[ni].sum = x * (rx - lx);
110         seg_data[ni].lazyAdd = 0;
111     }
112
113     void doPushMinEq(int ni, int lx, int rx, int x) {
114         if (seg_data[ni].mn >= x) doPushEq(ni, lx, rx, x);
115         else if (seg_data[ni].mx > x) {
116             if (seg_data[ni].secondMn == seg_data[ni].mx)
117                 seg_data[ni].secondMn = x;
118             seg_data[ni].sum -= (seg_data[ni].mx - x) * seg_data[ni].
119                 mxCnt;
120             seg_data[ni].mx = x;
121         }
122     }
123
124     void doPushMaxEq(int ni, int lx, int rx, int x) {
125         if (seg_data[ni].mx <= x) doPushEq(ni, lx, rx, x);
126         else if (seg_data[ni].mn < x) {
127             if (seg_data[ni].secondMx == seg_data[ni].mn)
128                 seg_data[ni].secondMx = x;
129
130             seg_data[ni].sum += (x - seg_data[ni].mn) * seg_data[ni].
131                 mnCnt;
132             seg_data[ni].mn = x;
133         }
134     }
135
136     void doPushSum(int ni, int lx, int rx, int x) {

```

```

134     if (seg_data[ni].mn == seg_data[ni].mx)
135         doPushEq(ni, lx, rx, seg_data[ni].mn + x);
136     else {
137         seg_data[ni].mx += x;
138         if (seg_data[ni].secondMx != -inf)
139             seg_data[ni].secondMx += x;
140
141         seg_data[ni].mn += x;
142         if (seg_data[ni].secondMn != inf)
143             seg_data[ni].secondMn += x;
144
145         seg_data[ni].sum += (rx - lx) * x;
146         seg_data[ni].lazyAdd += x;
147     }
148 }
149
150 void propagete(int ni, int lx, int rx) {
151     if (rx - lx == 1) return;
152
153     int mid = lx + (rx - lx) / 2;
154     if (seg_data[ni].isLazyEqual) {
155         doPushEq(2 * ni + 1, lx, mid, seg_data[ni].lazyEqual);
156         doPushEq(2 * ni + 2, mid, rx, seg_data[ni].lazyEqual);
157         seg_data[ni].lazyEqual = seg_data[ni].isLazyEqual = 0;
158     }
159     else {
160         doPushSum(2 * ni + 1, lx, mid, seg_data[ni].lazyAdd);
161         doPushSum(2 * ni + 2, mid, rx, seg_data[ni].lazyAdd);
162         seg_data[ni].lazyAdd = 0;
163
164         doPushMinEq(2 * ni + 1, lx, mid, seg_data[ni].mx);
165         doPushMinEq(2 * ni + 2, mid, rx, seg_data[ni].mx);
166
167         doPushMaxEq(2 * ni + 1, lx, mid, seg_data[ni].mn);
168         doPushMaxEq(2 * ni + 2, mid, rx, seg_data[ni].mn);
169     }
170 }
171
172 void minimize(int l, int r, int x, int ni, int lx, int rx) {
173     if (rx <= 1 || lx >= r || seg_data[ni].mx <= x)
174         return;
175     if (lx >= 1 && rx <= r && seg_data[ni].secondMx < x) {
176         doPushMinEq(ni, lx, rx, x);
177         return;
178     }
179
180     propagete(ni, lx, rx);
181
182     int mid = lx + (rx - lx) / 2;
183     minimize(l, r, x, 2 * ni + 1, lx, mid);
184     minimize(l, r, x, 2 * ni + 2, mid, rx);
185
186     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
187 2]);
188 }
189
190 void minimize(int l, int r, int x) {
191     minimize(l, r, x, 0, 0, tree_size);
192 }
193
194 void maximize(int l, int r, int x, int ni, int lx, int rx) {
195     if (rx <= 1 || lx >= r || seg_data[ni].mn >= x)
196         return;
197     if (lx >= 1 && rx <= r && seg_data[ni].secondMn > x) {
198         doPushMaxEq(ni, lx, rx, x);
199         return;
200     }
201
202     propagete(ni, lx, rx);
203
204     int mid = lx + (rx - lx) / 2;
205     maximize(l, r, x, 2 * ni + 1, lx, mid);
206     maximize(l, r, x, 2 * ni + 2, mid, rx);
207
208     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
209 2]);
210 }
211
212 void maximize(int l, int r, int x) {
213     maximize(l, r, x, 0, 0, tree_size);
214 }
215
216 void add(int l, int r, int x, int ni, int lx, int rx) {
217     if (rx <= 1 || lx >= r)
218         return;
219     if (lx >= 1 && rx <= r) {
220         doPushSum(ni, lx, rx, x);
221         return;
222     }
223
224     propagete(ni, lx, rx);
225
226     int mid = lx + (rx - lx) / 2;
227     add(l, r, x, 2 * ni + 1, lx, mid);
228     add(l, r, x, 2 * ni + 2, mid, rx);
229
230     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
231 2]);
232 }
233
234 void add(int l, int r, int x) {
235     add(l, r, x, 0, 0, tree_size);
236 }
237
238 void set(int l, int r, int x, int ni, int lx, int rx) {
239     if (rx <= 1 || lx >= r)
240         return;
241     if (lx >= 1 && rx <= r) {
242         doPushEq(ni, lx, rx, x);
243         return;
244     }
245
246     propagete(ni, lx, rx);

```

```

245     int mid = lx + (rx - lx) / 2;
246     set(l, r, x, 2 * ni + 1, lx, mid);
247     set(l, r, x, 2 * ni + 2, mid, rx);
248
249     seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
250 2]);
251 }
252
253 void set(int l, int r, int x) {
254     set(l, r, x, 0, 0, tree_size);
255 }
256
257 Node get_range(int l, int r, int ni, int lx, int rx) {
258     propagete(ni, lx, rx);
259
260     if (lx >= 1 && rx <= r)
261         return seg_data[ni];
262     if (rx <= 1 || lx >= r)
263         return Node();
264
265     int mid = (lx + rx) / 2;
266
267     Node lf = get_range(l, r, 2 * ni + 1, lx, mid);
268     Node ri = get_range(l, r, 2 * ni + 2, mid, rx);
269     return merge(lf, ri);
270 }
271
272 int get_sum(int l, int r) {
273     return get_range(l, r, 0, 0, tree_size).sum;
274 }
275
276 int get_mx(int l, int r) {
277     return get_range(l, r, 0, 0, tree_size).mx;
278 }
279
280 int get_mn(int l, int r) {
281     return get_range(l, r, 0, 0, tree_size).mn;
282 };

```

2.2 Binary Indexed Tree (BIT)

2.2.1 FenwickTree

```

1  template<typename T>
2  class FenwickTree {
3  public:
4      vector<T> tree;
5      int n;
6
7      // Default constructor
8      FenwickTree() : n(0) {}
9
10     // Constructor with size
11     FenwickTree(int n) {
12         init(n);
13     }
14
15     void init(int n) {
16         this->n = n;
17         // 1-based indexing needs size n + 1. Padding by 2 is safe.
18         tree.assign(n + 2, 0);
19     }
20
21     void update(int x, T val) {
22         x++; // Convert 0-based to 1-based
23         for (; x <= n; x += x & -x) {
24             tree[x] += val;
25         }
26     }
27
28     void assign(int x, T val) {
29         update(x, val - getRange(x, x));
30     }
31
32     T getPrefix(int x) {
33         x++; // Convert 0-based to 1-based
34         if (x <= 0) return 0;
35         T ret = 0;
36         for (; x; x -= x & -x) {
37             ret += tree[x];
38         }
39         return ret;
40     }
41
42     T getRange(int l, int r) {
43         if (l > r) return 0;
44         return getPrefix(r) - getPrefix(l - 1);
45     }
46
47     // Returns the 0-based index of the first prefix sum >= x
48     int lowerBound(T x) {
49         int pos = 0;
50         // sz > 0 is enough; don't break early if x == 0 to handle 0-
51         // value elements properly
52         for (int sz = (1 << __lg(n)); sz > 0; sz >= 1) {
53             if (pos + sz <= n && tree[pos + sz] < x) {
54                 x -= tree[pos + sz];
55                 pos += sz;
56             }
57         }
58         return pos; // pos is the 0-based index (since 1-based would be
59         pos + 1)
60     }
61 };

```

2.2.2 FenwickTree2D

```

1  template<typename T>
2  class FenwickTree2D {
3  public:
4      vector<vector<T>> tree;
5      int n, m;
6

```

```

7 // Default constructor
8 FenwickTree2D() : n(0), m(0) {}
9
10 // Constructor with dimensions
11 FenwickTree2D(int n, int m) {
12     init(n, m);
13 }
14
15 void init(int n, int m) {
16     this->n = n;
17     this->m = m;
18     // 1-based indexing needs sizes n + 1 and m + 1. Padding by 2
19     // is safe.
20     tree.assign(n + 2, vector<T>(m + 2, 0));
21 }
22
23 // Point update: Adds 'val' to the cell (x, y)
24 void update(int x, int y, T val) {
25     x++; y++; // Convert 0-based to 1-based
26     for (int i = x; i <= n; i += i & -i) {
27         for (int j = y; j <= m; j += j & -j) {
28             tree[i][j] += val;
29         }
30     }
31 }
32
33 // Point assignment: Sets the cell (x, y) to 'val'
34 void assign(int x, int y, T val) {
35     update(x, y, val - getRange(x, y, x, y));
36 }
37
38 // 2D Prefix sum: Returns the sum of the subgrid from (0, 0) to (x, y)
39 T getPrefix(int x, int y) {
40     x++; y++; // Convert 0-based to 1-based
41     if (x <= 0 || y <= 0) return 0;
42
43     // Clamp to maximum dimensions if queries exceed bounds
44     x = min(x, n);
45     y = min(y, m);
46
47     T ret = 0;
48     for (int i = x; i > 0; i -= i & -i) {
49         for (int j = y; j > 0; j -= j & -j) {
50             ret += tree[i][j];
51         }
52     }
53     return ret;
54 }
55
56 // 2D Range sum: Returns the sum of the subgrid from top-left (x1, y1) to bottom-right (x2, y2)
57 T getRange(int x1, int y1, int x2, int y2) {
58     if (x1 > x2 || y1 > y2) return 0;
59
60     // Inclusion-Exclusion Principle
61     return getPrefix(x2, y2)
62         - getPrefix(x1 - 1, y2)
63         - getPrefix(x2, y1 - 1)
64         + getPrefix(x1 - 1, y1 - 1);
65 }
66 };

```

2.2.3 Offline2DFenwickTree

```

1 template <typename T>
2 struct BIT2D {
3     int n;
4     vector<vector<int>> vals;
5     vector<vector<T>> bit;
6
7     int ind(const vector<int> &v, int x) {
8         return upper_bound(begin(v), end(v), x) - begin(v) - 1;
9     }
10
11     // n: the limit of the first dimension
12     // all update operations you will make
13     BIT2D(int n, vector<array<int, 2>> todo): n(n + 1), vals(n + 1),
14         bit(n + 1) {
15         sort(begin(todo), end(todo), [](auto &a, auto &b) { return a[1] < b[1]; });
16
17         for (int i = 0; i < n; i++) vals[i].push_back(-oo);
18         for (auto [r, c] : todo)
19             for (; r < n; r |= r + 1)
20                 if (vals[r].back() != c) vals[r].push_back(c);
21
22         for (int i = 0; i < n; i++) bit[i].resize(vals[i].size());
23     }
24
25     void add(int r, int c, T val) {
26         for (; r < n; r |= r + 1) {
27             int i = ind(vals[r], c);
28             for (; i < bit[r].size(); i |= i + 1) bit[r][i] += val;
29         }
30     }
31
32     T query(int r, int c) {
33         T ans = 0;
34         for (; r >= 0; r = (r & r + 1) - 1) {
35             int i = ind(vals[r], c);
36             for (; i >= 0; i = (i & i + 1) - 1) ans += bit[r][i];
37         }
38         return ans;
39     }
40
41     T query(int r1, int c1, int r2, int c2) {
42         return query(r2, c2) - query(r2, c1 - 1) - query(r1 - 1, c2) +
43             query(r1 - 1, c1 - 1);
44     }
45
46     void reset() {
47         for(auto &v: bit) for(auto &i: v) i = 0;
48     }
49 };

```

```

47 };

```

2.3 Sparse Table

2.3.1 SparseTable

```

1 #define sz(aa) (int)aa.size()
2
3 template <class T>
4 struct sparseTable {
5     vector<vector<T>> jmp;
6
7     void build(const vector<T>& V) {
8         jmp.assign(1, V);
9         for (int pw = 1, k = 1; pw * 2 <= sz(V); pw *= 2, ++k) {
10             jmp.emplace_back(sz(V) - pw * 2 + 1);
11             for (int j = 0; j < sz(jmp[k]); ++j) {
12                 jmp[k][j] = max(jmp[k - 1][j], jmp[k - 1][j + pw]);
13             }
14         }
15     }
16
17     T query(int l, int r) {
18         assert(l <= r);
19         int dep = 31 - __builtin_clz(r - l + 1);
20         return max(jmp[dep][l], jmp[dep][r - (1 << dep) + 1]);
21     }
22 };

```

2.3.2 SparseTable2D

```

1 #define sz(aa) (int)aa.size()
2
3 template <class T>
4 struct sparseTable2D {
5     int n, m;
6     // jmp[kr][kc][i][j] stores the max of a 2^kr by 2^kc rectangle
7     // starting at (i, j)
8     vector<vector<vector<vector<T>>>> jmp;
9
10     void build(const vector<vector<T>>& V) {
11         n = sz(V);
12         if (n == 0) return;
13         m = sz(V[0]);
14         if (m == 0) return;
15
16         int log_n = 32 - __builtin_clz(n);
17         int log_m = 32 - __builtin_clz(m);
18
19         jmp.assign(log_n, vector<vector<vector<T>>>(log_m, vector<
20             vector<T>>(n, vector<T>(m))));
21
22         // Base case: 1x1 rectangles (kr = 0, kc = 0)
23         for (int i = 0; i < n; ++i) {
24             for (int j = 0; j < m; ++j) {
25                 jmp[0][0][i][j] = V[i][j];
26             }
27         }
28
29         // Build 1D sparse tables for each row (kr = 0, increasing kc)
30         for (int kc = 1; kc < log_m; ++kc) {
31             for (int i = 0; i < n; ++i) {
32                 for (int j = 0; j + (1 << kc) <= m; ++j) {
33                     jmp[0][kc][i][j] = max(jmp[0][kc - 1][i][j],
34                         jmp[0][kc - 1][i][j + (1 <<
35                             (kc - 1))]);
36                 }
37             }
38         }
39
40         // Build the 2D tables by merging rows (increasing kr, for all kc)
41         for (int kr = 1; kr < log_n; ++kr) {
42             for (int kc = 0; kc < log_m; ++kc) {
43                 for (int i = 0; i + (1 << kr) <= n; ++i) {
44                     for (int j = 0; j + (1 << kc) <= m; ++j) {
45                         jmp[kr][kc][i][j] = max(jmp[kr - 1][kc][i][j],
46                             jmp[kr - 1][kc][i + (1 <<
47                                 (kr - 1))][j]);
48                     }
49                 }
50             }
51         }
52
53         // Queries the maximum in the subgrid from (r1, c1) to (r2, c2)
54         // inclusive
55         T query(int r1, int c1, int r2, int c2) {
56             assert(r1 <= r2 && c1 <= c2);
57             int kr = 31 - __builtin_clz(r2 - r1 + 1);
58             int kc = 31 - __builtin_clz(c2 - c1 + 1);
59
60             // Get the 4 overlapping sub-rectangles
61             T top_left = jmp[kr][kc][r1][c1];
62             T top_right = jmp[kr][kc][r1][c2 - (1 << kc) + 1];
63             T bottom_left = jmp[kr][kc][r2 - (1 << kr) + 1][c1];
64             T bottom_right = jmp[kr][kc][r2 - (1 << kr) + 1][c2 - (1 << kc) + 1];
65
66             return max({top_left, top_right, bottom_left, bottom_right});
67         }
68     };
69 };

```

2.3.3 SquareSparseTable

```

1 #define sz(aa) (int)aa.size()
2
3 template <class T>
4 struct squareSparseTable {
5     int n, m;
6     // jmp[k][i][j] stores the max of a 2^k by 2^k square starting at (
7     // i, j)

```

```

7   vector<vector<vector<T>>> jmp;
8
9   void build(const vector<vector<T>>& V) {
10       n = sz(V);
11       if (n == 0) return;
12       m = sz(V[0]);
13       if (m == 0) return;
14
15       // The mazimum possible square side length is min(n, m)
16       int max_len = min(n, m);
17       int log_max = 32 - __builtin_clz(max_len);
18
19       // Only 3 dimensions: log(min(N,M)) x N x M
20       jmp.assign(log_max, vector<vector<T>>(n, vector<T>(m)));
21
22       // Base case: 1x1 squares (k = 0)
23       for (int i = 0; i < n; ++i) {
24           for (int j = 0; j < m; ++j) {
25               jmp[0][i][j] = V[i][j];
26           }
27       }
28
29       // Build larger squares by combining 4 smaller squares
30       for (int k = 1; k < log_max; ++k) {
31           int pw = 1 << (k - 1); // Length of the smaller squares
32                                   (2^(k-1))
33           for (int i = 0; i + (pw * 2) <= n; ++i) {
34               for (int j = 0; j + (pw * 2) <= m; ++j) {
35                   jmp[k][i][j] = max({
36                       jmp[k - 1][i][j],           // Top-Left
37                       jmp[k - 1][i][j + pw],       // Top-Right
38                       jmp[k - 1][i + pw][j],       // Bottom-Left
39                       jmp[k - 1][i + pw][j + pw]    // Bottom-Right
40                   });
41               }
42           }
43       }
44
45       // Queries the mazimum in a square from (r1, c1) to (r2, c2)
46       T query(int r1, int c1, int r2, int c2) {
47           assert(r1 <= r2 && c1 <= c2);
48           assert((r2 - r1) == (c2 - c1)); // Ensure it is actually a
49                                           square
50
51           int len = r2 - r1 + 1;
52           int k = 31 - __builtin_clz(len);
53           int pw = 1 << k;
54
55           // Overlap four 2^k x 2^k squares to cover the arbitrary S x S
56           // square
57           return max({
58               jmp[k][r1][c1],
59               jmp[k][r1][c2 - pw + 1],
60               jmp[k][r2 - pw + 1][c1],
61               jmp[k][r2 - pw + 1][c2 - pw + 1]
62           });
63 };
```

2.4 Trie

2.4.1 StringTrie

```

1  struct Trie {
2      vector<vector<int>>> trie;
3      vector<int> cnt;
4      // vector<int>leaves;
5      int m, sz;
6
7      int addNode() {
8          trie.emplace_back(m, -1);
9          cnt.emplace_back();
10         // leaves.emplace_back();
11         sz++;
12         return sz - 1;
13     }
14
15     Trie(int m = 26) : m(m), sz(0) {
16         addNode();
17     };
18
19     // insert or remove
20     void insert(string &s, int type = 1) {
21         int cur = 0;
22         for (auto c: s) {
23             if (trie[cur][c - 'a'] == -1)
24                 trie[cur][c - 'a'] = addNode();
25             cur = trie[cur][c - 'a'];
26             cnt[cur] += type;
27         }
28         // leaves[cur] += type;
29     }
30
31
32     int query(string &s) {
33         int cur = 0;
34         for (auto c: s) {
35             if (trie[cur][c - 'a'] == -1)
36                 return 0;
37             cur = trie[cur][c - 'a'];
38         }
39         return cnt[cur];
40     }
41 };
```

2.4.2 BinaryTrie

```

1  struct Trie{
2      vector<vector<int>>>trie;
3      vector<int>cnt;
4      // vector<int>leaves;
5      int mxBit,sz;
6
```

```

7      int addNode(){
8          trie.emplace_back(2,-1);
9          cnt.emplace_back();
10         // leaves.emplace_back();
11         sz++;
12         return sz - 1;
13     }
14
15     Trie(int mx = 60): mxBit(mx),sz(0){
16         addNode();
17     };
18
19     // insert or remove
20     void insert(ll x,int type = 1){
21         int cur = 0;
22         cnt[cur] += type;
23         for (int i = mxBit; i >= 0; --i) {
24             int t = (x >> i)&1;
25             if(trie[cur][t] == -1)
26                 trie[cur][t] = addNode();
27             cur = trie[cur][t];
28             cnt[cur] += type;
29         }
30         // leaves[cur] += type;
31     }
32
33     ll maxXor(ll x){
34         // no elements in trie
35         int cur = 0;
36         if(!cnt[cur])return -1e9;
37         for (int i = mxBit; i >= 0; --i) {
38             int t = (x >> i)&1^1;
39             if(trie[cur][t] == -1 || !cnt[trie[cur][t]])t ^= 1;
40             cur = trie[cur][t];
41             if(t)x ^= 1ll << i;
42         }
43         return x;
44     }
45 };
```

2.5 Disjoint Set Union (DSU)

2.5.1 DSU

```

1  struct DSU {
2      vector<int> par, sz;
3
4      DSU(int n) : par(n), sz(n, 1) { iota(par.begin(), par.end(), 0); }
5
6      int find(int x) {
7          if(x == par[x])return x;
8          return par[x] = find(par[x]);
9      }
10
11     bool same(int x, int y) { return find(x) == find(y); }
12
13     bool join(int x, int y) {
14         x = find(x);
15         y = find(y);
16         if (x == y) return false;
17         if (sz[x] < sz[y])
18             swap(x, y);
19         sz[x] += sz[y];
20         par[y] = x;
21         return true;
22     }
23
24     int size(int x) { return sz[find(x)]; }
25 };
```

2.5.2 DSURollback

```

1  struct DSURollback {
2      vector<int> par, sz;
3      vector<pair<int, int>> history; // Stores {child_node,
4                                      old_size_of_parent}
5
6      DSURollback(int n) : par(n), sz(n, 1) {
7          iota(par.begin(), par.end(), 0);
8      }
9
10     // Path compression is removed to preserve tree structure for
11     // rollbacks
12     int find(int x) {
13         if(x == par[x]) return x;
14         return find(par[x]);
15     }
16
17     bool same(int x, int y) { return find(x) == find(y); }
18
19     bool join(int x, int y) {
20         x = find(x);
21         y = find(y);
22
23         if (x == y) {
24             // Push dummy state to keep the rollback count 1:1 with
25             // join calls
26             history.push_back({-1, -1});
27             return false;
28         }
29
30         if (sz[x] < sz[y]) swap(x, y);
31
32         // Record the node that becomes the child and the parent's
33         // current size
34         history.push_back({y, sz[x]});
35
36         sz[x] += sz[y];
37         par[y] = x;
38         return true;
39     }
40
41     int size(int x) { return sz[find(x)]; }
42 };
```



```

39 // Undoes the last join operation
40 void rollback() {
41     if (history.empty()) return;
42
43     auto [y, old_sz_x] = history.back();
44     history.pop_back();
45
46     if (y != -1) {
47         int x = par[y];
48         sz[x] = old_sz_x;
49         par[y] = y; // Restore the child to be its own parent
50     }
51 }
52
53 // Returns the current number of operations performed
54 int get_state() {
55     return history.size();
56 }
57
58 // Rolls back to a specific saved state
59 void rollback_to(int state) {
60     while (history.size() > state) {
61         rollback();
62     }
63 }
64 };

```

2.6 Square Root Decomposition

2.6.1 MoAlgorithm

```

1 class MoArray {
2 private:
3     struct Query {
4         int l, r, idx;
5         int bnd;
6
7         bool operator<(const Query& other) const {
8             if (bnd != other.bnd)
9                 return bnd < other.bnd;
10            return (bnd & 1) ? (r < other.r) : (r > other.r);
11        }
12    };
13
14    int n;
15    int block_size;
16    vector<int> a;
17
18    // --- PROBLEM SPECIFIC STATE ---
19    long long ans;
20    vector<int> freq;
21
22    inline void add(int idx) {
23        if (freq[a[idx]] == 0) ans++;
24        freq[a[idx]]++;
25    }
26
27    inline void remove(int idx) {
28        freq[a[idx]]--;
29        if (freq[a[idx]] == 0) ans--;
30    }
31
32    inline long long get_answer() const {
33        return ans;
34    }
35    // -----
36
37 public:
38     MoArray(const vector<int>& arr) : n(arr.size()), a(arr) {
39         int max_val = 0;
40         for (int x : a) {
41             max_val = max(max_val, x);
42         }
43         freq.assign(max_val + 1, 0);
44         ans = 0;
45     }
46
47     vector<long long> solve(const vector<pair<int, int>>& raw_queries) {
48         int q = raw_queries.size();
49         if (q == 0) return {};
50
51         block_size = max(1, (int)(n / sqrt(q)));
52         vector<Query> queries(q);
53
54         for (int i = 0; i < q; ++i) {
55             queries[i].l = raw_queries[i].first;
56             queries[i].r = raw_queries[i].second;
57             queries[i].idx = i;
58             queries[i].bnd = queries[i].l / block_size;
59         }
60
61         sort(queries.begin(), queries.end());
62
63         vector<long long> answers(q);
64         int cur_l = 0, cur_r = -1;
65
66         for (const Query& qry : queries) {
67             while (cur_l > qry.l) add(--cur_l);
68             while (cur_r < qry.r) add(++cur_r);
69             while (cur_l < qry.l) remove(cur_l++);
70             while (cur_r > qry.r) remove(cur_r--);
71             answers[qry.idx] = get_answer();
72         }
73
74         return answers;
75     }
76 };

```

2.6.2 MoOnSubtrees

```

1 class MoSubtree {
2 private:
3     struct Query {

```

```

4         int l, r, idx;
5         int bnd;
6
7         bool operator<(const Query& other) const {
8             if (bnd != other.bnd) return bnd < other.bnd;
9             return (bnd & 1) ? (r < other.r) : (r > other.r);
10        }
11    };
12
13    int n;
14    int block_size;
15    vector<vector<int>> adj;
16    vector<int> in, out, flat, clr;
17
18    // --- PROBLEM SPECIFIC STATE ---
19    int mx_freq;
20    vector<int> freq;
21    vector<long long> sum_freq;
22
23    inline void add(int node_idx) {
24        int c = clr[node_idx];
25        sum_freq[freq[c]] -= c;
26        freq[c]++;
27        sum_freq[freq[c]] += c;
28        if (freq[c] > mx_freq) mx_freq = freq[c];
29    }
30
31    inline void remove(int node_idx) {
32        int c = clr[node_idx];
33        sum_freq[freq[c]] -= c;
34        freq[c]--;
35        sum_freq[freq[c]] += c;
36        while (mx_freq > 0 && sum_freq[mx_freq] == 0) mx_freq--;
37    }
38
39    inline long long get_answer() const {
40        return sum_freq[mx_freq];
41    }
42    // -----
43
44    void dfs(int u, int p) {
45        in[u] = flat.size();
46        flat.push_back(u);
47        for (int v : adj[u]) {
48            if (v == p) continue;
49            dfs(v, u);
50        }
51        out[u] = flat.size() - 1;
52    }
53
54 public:
55     MoSubtree(int n, int max_val, const vector<int>& colors) {
56         n(n), adj(n + 1), in(n + 1), out(n + 1), clr(colors) {
57             mx_freq = 0;
58             freq.assign(max_val + 1, 0);
59             sum_freq.assign(n + 1, 0);
60         }
61
62     void add_edge(int u, int v) {
63         adj[u].push_back(v);
64         adj[v].push_back(u);
65     }
66
67     vector<long long> solve(int root, const vector<int>& query_nodes) {
68         flat.reserve(n);
69         dfs(root, 0);
70
71         int q = query_nodes.size();
72         if (q == 0) return {};
73
74         block_size = max(1, (int)(n / sqrt(q)));
75         vector<Query> queries(q);
76
77         for (int i = 0; i < q; ++i) {
78             int u = query_nodes[i];
79             queries[i].l = in[u];
80             queries[i].r = out[u];
81             queries[i].idx = i;
82             queries[i].bnd = queries[i].l / block_size;
83         }
84
85         sort(queries.begin(), queries.end());
86
87         vector<long long> answers(q);
88         int cur_l = 0, cur_r = -1;
89
90         for (const Query& qry : queries) {
91             while (cur_l > qry.l) add(flat[--cur_l]);
92             while (cur_r < qry.r) add(flat[++cur_r]);
93             while (cur_l < qry.l) remove(flat[cur_l++]);
94             while (cur_r > qry.r) remove(flat[cur_r--]);
95             answers[qry.idx] = get_answer();
96         }
97
98         return answers;
99     }
100 };

```

2.6.3 MoOnPaths

```

1 class MoTreePath {
2 private:
3     int n, LOG, block_size;
4     vector<vector<int>> adj, anc;
5     vector<int> depth, in, out, flat;
6     vector<long long> up, a;
7     vector<bool> exist;
8
9     // --- PROBLEM SPECIFIC STATE ---
10    long long ans;
11    vector<deque<int>> dep; // Tracks nodes by depth
12
13    void add(int idx) {
14        if (!dep[depth[idx]].empty()) {

```



```

15         ans += 1ll * a[dep[depth[idx]].back()] * a[idx];
16     }
17     dep[depth[idx]].push_back(idx);
18 }
19
20 void remove(int idx) {
21     if (dep[depth[idx]].size() == 2) {
22         ans -= a[*dep[depth[idx]].begin()] * a[*next(dep[depth[idx]].begin())];
23     }
24     if (dep[depth[idx]].front() == idx) {
25         dep[depth[idx]].pop_front();
26     } else {
27         dep[depth[idx]].pop_back();
28     }
29 }
30
31 void update(int idx) {
32     int node = flat[idx];
33     if (exist[node]) {
34         remove(node);
35     } else {
36         add(node);
37     }
38     exist[node] = !exist[node];
39 }
40
41 inline long long get_answer() const {
42     return ans;
43 }
44 // -----
45
46 void dfs(int u, int p = -1) {
47     if (p != -1) {
48         anc[u][0] = p;
49         up[u] = a[u] * a[u] + up[p];
50     } else {
51         depth[u] = 0;
52         up[u] = a[u] * a[u];
53     }
54     in[u] = flat.size();
55     flat.push_back(u);
56
57     for (int v : adj[u]) {
58         if (v == p) continue;
59         depth[v] = depth[u] + 1;
60         dfs(v, u);
61     }
62
63     out[u] = flat.size();
64     flat.push_back(u); // Push again for Eulerian tour
65 }
66
67 void build_lca() {
68     for (int k = 1; k < LOG; k++) {
69         for (int i = 1; i <= n; i++) {
70             anc[i][k] = anc[anc[i][k - 1]][k - 1];
71         }
72     }
73 }
74
75 int lift(int x, int k) {
76     for (int bit = 0; bit < LOG; bit++) {
77         if ((1 << bit) & k) x = anc[x][bit];
78     }
79     return x;
80 }
81
82 int lca(int u, int v) {
83     if (depth[u] < depth[v]) swap(u, v);
84     u = lift(u, depth[u] - depth[v]);
85     if (u == v) return u;
86     for (int bit = LOG - 1; bit >= 0; bit--) {
87         if (anc[u][bit] != anc[v][bit]) {
88             u = anc[u][bit];
89             v = anc[v][bit];
90         }
91     }
92     return anc[u][0];
93 }
94
95 public:
96     struct Query {
97         int l, r, idx, x = -1;
98         int bnd;
99
100         bool operator<(const Query& other) const {
101             if (bnd != other.bnd) return bnd < other.bnd;
102             return (bnd & 1) ? (r < other.r) : (r > other.r);
103         }
104     };
105
106     MoTreePath(int n_nodes, const vector<long long>& node_values)
107         : n(n_nodes), a(node_values), LOG(log2(n_nodes) + 2) {
108
109         adj.assign(n + 1, vector<int>());
110         anc.assign(n + 1, vector<int>(LOG, 0));
111         depth.assign(n + 1, 0);
112         in.assign(n + 1, 0);
113         out.assign(n + 1, 0);
114         up.assign(n + 1, 0);
115         exist.assign(n + 1, false);
116         dep.assign(n + 1, deque<int>());
117         ans = 0;
118     }
119
120     void add_edge(int u, int v) {
121         adj[u].push_back(v);
122         adj[v].push_back(u);
123     }
124
125     vector<long long> solve(int root, vector<Query>& queries) {
126         flat.clear();
127         dfs(root);

```

```

128         build_lca();
129
130         int q = queries.size();
131         if (q == 0) return {};
132
133         block_size = max(1, (int)(flat.size() / sqrt(q)));
134
135         for (auto& qry : queries) {
136             qry.bnd = qry.l / block_size;
137         }
138
139         sort(queries.begin(), queries.end());
140
141         vector<long long> answers(q);
142         int cur_l = 0, cur_r = -1;
143
144         for (const Query& qry : queries) {
145             while (cur_l < qry.l) update(cur_l++);
146             while (cur_r > qry.r) update(cur_r--);
147             while (cur_l > qry.l) update(--cur_l);
148             while (cur_r < qry.r) update(++cur_r);
149
150             answers[qry.idx] = get_answer() + (qry.x != -1 ? up[qry.x] : 0);
151         }
152         return answers;
153     }
154 };

```

2.7 Balanced Binary Search Tree (BBST)

2.7.1 Treap

```

1 mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
2
3 template <typename T>
4 struct Treap {
5     struct Node {
6         T key, val, sum;
7         T lazy = 0; // 1. Added lazy tag
8         int sz = 1;
9         uint64_t p = rng();
10        Node *l = nullptr, *r = nullptr;
11
12        Node(T k, T v = T()) : key(k), val(v), sum(v) {}
13    } *root = nullptr;
14
15    ~Treap() { destroy(root); }
16
17    void destroy(Node* t) {
18        if (!t) return;
19        destroy(t->l);
20        destroy(t->r);
21        delete t;
22    }
23
24    int sz(Node* t) { return t ? t->sz : 0; }
25    T sub_val(Node* t) { return t ? t->sum : 0; }
26
27    // 2. The Push Function: Propagates pending updates to children
28    void push(Node* t) {
29        if (!t || t->lazy == 0) return;
30
31        if (t->l) {
32            t->l->key += t->lazy; // Shift the key (critical for the DP solution)
33            t->l->val += t->lazy; // Shift the value
34            t->l->sum += t->lazy * sz(t->l); // Update subtree sum
35            t->l->lazy += t->lazy; // Pass the tag down
36        }
37        if (t->r) {
38            t->r->key += t->lazy;
39            t->r->val += t->lazy;
40            t->r->sum += t->lazy * sz(t->r);
41            t->r->lazy += t->lazy;
42        }
43        t->lazy = 0; // Clear the tag on the current node
44    }
45
46    Node* update(Node* t) {
47        if (!t) return nullptr;
48        t->sz = 1 + sz(t->l) + sz(t->r);
49        t->sum = t->val + sub_val(t->l) + sub_val(t->r);
50        return t;
51    }
52
53    Node* merge(Node* l, Node* r) {
54        push(l); // 3. Push before making routing decisions
55        push(r);
56        if (!l || !r) return l ? l : r;
57        if (l->p > r->p) {
58            l->r = merge(l->r, r);
59            return update(l);
60        }
61        r->l = merge(l, r->l);
62        return update(r);
63    }
64
65    pair<Node*, Node*> split(Node* t, T k) {
66        if (!t) return {nullptr, nullptr};
67        push(t); // 3. Push before evaluating keys for splits
68        if (t->key <= k) {
69            auto [l, r] = split(t->r, k);
70            t->r = l;
71            return {update(t), r};
72        }
73        auto [l, r] = split(t->l, k);
74        t->l = r;
75        return {l, update(t)};
76    }
77
78    // --- Core Operations ---
79
80

```

```

81 // Range Update Method
82 void add_range(T l_val, T r_val, T add_val) {
83     if (l_val > r_val) return;
84
85     auto [left_mid, right] = split(root, r_val);
86     auto [left, mid] = split(left_mid, l_val - 1);
87
88     // Apply the lazy tag to the entire isolated segment
89     if (mid) {
90         mid->key += add_val;
91         mid->val += add_val;
92         mid->sum += add_val * sz(mid);
93         mid->lazy += add_val;
94     }
95
96     root = merge(merge(left, mid), right);
97 }
98
99 bool search(T x) {
100     Node* curr = root;
101     while (curr) {
102         push(curr); // Push during traversal
103         if (curr->key == x) return true;
104         curr = (x < curr->key) ? curr->l : curr->r;
105     }
106     return false;
107 }
108
109 void insert(T x, T val = T()) {
110     auto [l, r] = split(root, x);
111     auto [l2, r2] = split(l, x - 1);
112     if (!r2) {
113         r2 = new Node(x, val);
114     } else {
115         push(r2); // Push before modifying
116         r2->val += val;
117         update(r2);
118     }
119     root = merge(merge(l2, r2), r);
120 }
121
122 void erase(T x) {
123     auto [l, r] = split(root, x);
124     auto [l2, r2] = split(l, x - 1);
125     destroy(r2);
126     root = merge(l2, r);
127 }
128
129 // --- Utility Operations ---
130
131 Node* find_by_order(Node* t, int k) {
132     if (!t) return nullptr;
133     push(t); // Push before accessing children sizes
134     int left_sz = sz(t->l);
135     if (k < left_sz) return find_by_order(t->l, k);
136     if (k == left_sz) return t;
137     return find_by_order(t->r, k - left_sz - 1);
138 }
139 Node* find_by_order(int k) { return find_by_order(root, k); }
140
141 int order_of_key(Node* t, T k) {
142     if (!t) return 0;
143     push(t); // Push before comparing keys
144     if (t->key >= k) return order_of_key(t->l, k);
145     return sz(t->l) + 1 + order_of_key(t->r, k);
146 }
147 int order_of_key(T k) { return order_of_key(root, k); }
148
149 void print(Node* t) {
150     if (!t) return;
151     push(t); // Push before printing to get true values
152     print(t->l);
153     cout << t->key << " ";
154     print(t->r);
155 }
156 void print() {
157     print(root);
158     cout << '\n';
159 }
160 };

```

2.7.2 ImplicitTreap

```

1 mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
2
3 template <typename T>
4 struct ImplicitTreap {
5     struct Node {
6         T val, sum;
7         T lazy = 0; // Lazy tag for range additions
8         bool rev = false; // Lazy tag for range reversals
9         int sz = 1;
10        uint64_t p = rng();
11        Node *l = nullptr, *r = nullptr;
12
13        Node(T v = T()) : val(v), sum(v) {}
14    } * root = nullptr;
15
16    // Default Constructor
17    ImplicitTreap() = default;
18
19    // Vector Constructor - Builds treap in O(N) time
20    ImplicitTreap(const vector<T>& a) {
21        build(a);
22    }
23
24    ~ImplicitTreap() { destroy(root); }
25
26    void destroy(Node* t) {
27        if (!t) return;
28        destroy(t->l);
29        destroy(t->r);
30        delete t;
31    }

```

```

32 int sz(Node* t) { return t ? t->sz : 0; }
33 T sub_val(Node* t) { return t ? t->sum : 0; }
34
35 // Propagates pending lazy updates to children
36 void push(Node* t) {
37     if (!t) return;
38
39     // 1. Process pending reversal
40     if (t->rev) {
41         swap(t->l, t->r);
42         if (t->l) t->l->rev ^= true;
43         if (t->r) t->r->rev ^= true;
44         t->rev = false;
45     }
46
47     // 2. Process pending value addition
48     if (t->lazy != 0) {
49         if (t->l) {
50             t->l->val += t->lazy;
51             t->l->sum += t->lazy * sz(t->l);
52             t->l->lazy += t->lazy;
53         }
54         if (t->r) {
55             t->r->val += t->lazy;
56             t->r->sum += t->lazy * sz(t->r);
57             t->r->lazy += t->lazy;
58         }
59         t->lazy = 0; // Clear the tag
60     }
61 }
62
63 Node* update(Node* t) {
64     if (t) {
65         t->sz = 1 + sz(t->l) + sz(t->r);
66         t->sum = t->val + sub_val(t->l) + sub_val(t->r);
67     }
68     return t;
69 }
70
71 // --- O(N) Linear Time Build ---
72 void build(const vector<T>& a) {
73     destroy(root);
74     root = nullptr;
75     if (a.empty()) return;
76
77     vector<Node*> st;
78     for (const T& val : a) {
79         Node* curr = new Node(val);
80         Node* last = nullptr;
81
82         // Maintain max-heap property on random priority 'p'
83         while (!st.empty() && st.back()->p < curr->p) {
84             last = st.back();
85             st.pop_back();
86         }
87
88         curr->l = last;
89         if (!st.empty()) {
90             st.back()->r = curr;
91         }
92         st.push_back(curr);
93     }
94
95     // Post-order pass to compute subtree sizes and sums in O(N)
96     auto recompute = [&](auto& self, Node* t) -> void {
97         if (!t) return;
98         self(self, t->l);
99         self(self, t->r);
100         update(t);
101     };
102
103     root = st.front(); // Bottom of stack is the root
104     recompute(recompute, root);
105
106 // --- Dynamic Array Operations ---
107
108 Node* merge(Node* l, Node* r) {
109     push(l);
110     push(r);
111     if (!l || !r) return l ? l : r;
112     if (l->p > r->p) {
113         l->r = merge(l->r, r);
114         return update(l);
115     }
116     r->l = merge(l, r->l);
117     return update(r);
118 }
119
120 pair<Node*, Node*> split(Node* t, int k) {
121     if (!t) return {nullptr, nullptr};
122     push(t);
123
124     int left_sz = sz(t->l);
125     if (left_sz < k) {
126         auto [l, r] = split(t->r, k - left_sz - 1);
127         t->r = l;
128         return {update(t), r};
129     } else {
130         auto [l, r] = split(t->l, k);
131         t->l = r;
132         return {l, update(t)};
133     }
134 }
135
136 int size() { return sz(root); }
137
138 void insert(int idx, T val) {
139     auto [l, r] = split(root, idx);
140     Node* node = new Node(val);
141     root = merge(merge(l, node), r);
142 }
143
144 }
145

```

```

146 void erase(int idx) {
147     if (idx < 0 || idx >= sz(root)) return;
148     auto [left_mid, right] = split(root, idx + 1);
149     auto [left, mid] = split(left_mid, idx);
150     destroy(mid);
151     root = merge(left, right);
152 }
153
154 void reverse_range(int l, int r) {
155     if (l >= r || l < 0 || r >= sz(root)) return;
156
157     auto [left_mid, right] = split(root, r + 1);
158     auto [left, mid] = split(left_mid, l);
159
160     if (mid) mid->rev ^= true;
161
162     root = merge(merge(left, mid), right);
163 }
164
165 void add_range(int l, int r, T add_val) {
166     if (l > r || l < 0 || r >= sz(root)) return;
167
168     auto [left_mid, right] = split(root, r + 1);
169     auto [left, mid] = split(left_mid, l);
170
171     if (mid) {
172         mid->val += add_val;
173         mid->sum += add_val * sz(mid);
174         mid->lazy += add_val;
175     }
176
177     root = merge(merge(left, mid), right);
178 }
179
180 T query_range(int l, int r) {
181     if (l > r || l < 0 || r >= sz(root)) return T();
182
183     auto [left_mid, right] = split(root, r + 1);
184     auto [left, mid] = split(left_mid, l);
185
186     T ans = sub_val(mid);
187
188     root = merge(merge(left, mid), right);
189     return ans;
190 }
191
192 T get(int idx) {
193     return query_range(idx, idx);
194 }
195
196 void print(Node* t) {
197     if (!t) return;
198     push(t);
199     print(t->l);
200     cout << t->val << " ";
201     print(t->r);
202 }
203
204 void print() {
205     print(root);
206     cout << '\n';
207 }
208 };

```

2.8 Policy Based Data Structures

2.8.1 OrderedSet

```

1 #include <ext/pb_ds/assoc_container.hpp>
2 #include <ext/pb_ds/tree_policy.hpp>
3
4 using namespace __gnu_pbds;
5 template<class t> using ordered_set = tree<t, null_type, less<t>,
    rb_tree_tag, tree_order_statistics_node_update>;

```

2.9 Monotonic Data Structures

2.9.1 TwoStackQueue

```

1 struct TwoStackQ {
2     struct Node {
3         ll mx = -llinf, mn = llinf, val;
4
5         Node(): val(0) {
6         }
7
8         Node(ll x): mx(x), mn(x), val(x) {
9         }
10    };
11
12    stack<Node> a, b;
13
14    int size() { return a.size() + b.size(); }
15
16    void mrg(Node& a, Node& b) {
17        a.mn = min(a.mn, b.mn);
18        a.mx = max(a.mx, b.mx);
19    }
20
21    void push(ll val) {
22        auto nd = Node(val);
23        if (!a.empty()) mrg(nd, a.top());
24        a.push(nd);
25    }
26
27    void move() {
28        while (!a.empty()) {
29            auto nd = Node(a.top().val);
30            if (!b.empty()) mrg(nd, b.top());
31            b.push(nd), a.pop();
32        }
33    }
34
35    Node get() {

```

```

36        Node res;
37        if (!b.empty()) mrg(res, b.top());
38        if (!a.empty()) mrg(res, a.top());
39        return res;
40    }
41
42    void pop() {
43        if (b.empty()) move();
44        if (!b.empty()) b.pop();
45    }
46 };

```

3 Graph Theory

3.1 Graph Traversal

3.1.1 TopoSort

```

1 struct TopoSort {
2     int n;
3     vector<vector<int>> adj;
4     vector<int> in_degree, order;
5
6     TopoSort(int n) : n(n), adj(n + 1), in_degree(n + 1, 0) {}
7
8     void add_edge(int u, int v) {
9         adj[u].push_back(v);
10        in_degree[v]++;
11    }
12
13    bool solve() {
14        queue<int> q;
15        for (int i = 1; i <= n; ++i) {
16            if (in_degree[i] == 0) q.push(i);
17        }
18
19        while (!q.empty()) {
20            int u = q.front();
21            q.pop();
22            order.push_back(u);
23
24            for (int v : adj[u]) {
25                if (--in_degree[v] == 0) {
26                    q.push(v);
27                }
28            }
29        }
30        return order.size() == n; // Returns false if there is a cycle
31    }
32 };

```

3.2 Shortest Paths

3.2.1 Dijkstra

```

1 struct Dijkstra {
2     int n;
3     vector<vector<pair<int, ll>>> adj;
4     vector<ll> dist;
5     vector<int> parent;
6
7     Dijkstra(int n) : n(n), adj(n + 1), dist(n + 1, inf), parent(n + 1,
        -1) {}
8
9     void add_edge(int u, int v, ll w, bool directed = false) {
10        adj[u].push_back({v, w});
11        if (!directed) adj[v].push_back({u, w});
12    }
13
14    void solve(int source) {
15        fill(dist.begin(), dist.end(), inf);
16        fill(parent.begin(), parent.end(), -1);
17        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<
            pair<ll, int>>> pq;
18
19        dist[source] = 0;
20        pq.push({0, source});
21
22        while (!pq.empty()) {
23            auto [d, u] = pq.top();
24            pq.pop();
25
26            if (d > dist[u]) continue;
27
28            for (auto& [v, w] : adj[u]) {
29                if (dist[u] + w < dist[v]) {
30                    dist[v] = dist[u] + w;
31                    parent[v] = u;
32                    pq.push({dist[v], v});
33                }
34            }
35        }
36    }
37 };

```

3.2.2 FloydWarshall

```

1 struct FloydWarshall {
2     int n;
3     vector<vector<ll>> dist;
4
5     FloydWarshall(int n) : n(n), dist(n + 1, vector<ll>(n + 1, inf)) {
6         for (int i = 0; i <= n; ++i) dist[i][i] = 0;
7     }
8
9     void add_edge(int u, int v, ll w, bool directed = false) {
10        dist[u][v] = min(dist[u][v], w);
11        if (!directed) dist[v][u] = min(dist[v][u], w);
12    }
13
14    void solve() {
15        for (int k = 1; k <= n; ++k) {
16            for (int i = 1; i <= n; ++i) {

```

```

17         for (int j = 1; j <= n; ++j) {
18             if (dist[i][k] < inf && dist[k][j] < inf) {
19                 dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
20             }
21         }
22     }
23 }
24 }
25 };

```

3.3 Graph Connectivity

3.3.1 BridgesAndCutEdges

```

1 int n;
2 vector<int> adj[N];
3
4 bool vis[N];
5 int in[N], low[N];
6 int timer;
7
8 set<pair<int, int>> bridges;
9
10 void dfs(int u, int p = 0) {
11     vis[u] = true;
12     in[u] = low[u] = timer++;
13     for (int v: adj[u]) {
14         if (v == p) continue;
15         if (vis[v]) {
16             low[u] = min(low[u], in[v]);
17         } else {
18             dfs(v, u);
19             low[u] = min(low[u], low[v]);
20             if (low[v] > in[u]) {
21                 /// (u, v) is a Bridge
22                 bridges.insert({u, v});
23                 bridges.insert({v, u});
24             }
25         }
26     }
27 }
28
29 void find_bridges() {
30     timer = 0;
31     fill(vis, vis + n + 1, false);
32     for (int i = 1; i <= n; ++i) {
33         if (!vis[i])
34             dfs(i);
35     }
36 }

```

3.3.2 ArticulationPoints

```

1 int n;
2 vector<int> adj[N];
3
4 bool vis[N];
5 int in[N], low[N];
6 int timer;
7
8 bool cutPoint[N];
9
10 void dfs(int u, int p = -1) {
11     vis[u] = true;
12     in[u] = low[u] = timer++;
13     int children = 0;
14     for (int v: adj[u]) {
15         if (v == p) continue;
16         if (vis[v]) {
17             low[u] = min(low[u], in[v]);
18         } else {
19             dfs(v, u);
20             low[u] = min(low[u], low[v]);
21             if (low[v] >= in[u] && p != -1) {
22                 /// u is a Cut Point
23                 cutPoint[u] = true;
24             }
25             children++;
26         }
27     }
28     if (p == -1 && children > 1) {
29         /// u is a Cut Point
30         cutPoint[u] = true;
31     }
32 }
33
34 bool connected = true;
35
36 void find_cut_points() {
37     timer = 0;
38     fill(vis, vis + n + 1, false);
39     for (int i = 1; i <= n; ++i) {
40         if (!vis[i]) {
41             if (i != 1)
42                 connected = false;
43             dfs(i);
44         }
45     }
46 }

```

3.3.3 SCC_Tarjan

```

1 /// assuming nodes are zero based
2 struct SCC {
3     vector<vector<int>> adj, adjRev, comps;
4     vector<pair<int, int>> edges;
5     vector<int> revOut, compOf;
6     vector<bool> vis;
7     int N;
8
9     void init(int n) {
10         N = n;

```

```

11         adj.resize(n);
12         adjRev.resize(n);
13         vis.resize(n);
14         compOf.resize(n);
15     }
16
17     void addEdge(int u, int v) {
18         edges.pb(make_pair(u, v));
19         adj[u].pb(v);
20         adjRev[v].pb(u);
21     }
22
23     void dfs1(int u) {
24         vis[u] = true;
25         for (auto v: adj[u])
26             if (!vis[v])
27                 dfs1(v);
28         revOut.pb(u);
29     }
30
31     void dfs2(int u) {
32         vis[u] = true;
33         comps.back().pb(u);
34         compOf[u] = comps.size() - 1;
35         for (auto v: adjRev[u])
36             if (!vis[v]) dfs2(v);
37     }
38
39     void gen() {
40         fill(all(vis), false);
41         for (int i = 0; i < N; ++i) {
42             if (!vis[i])
43                 dfs1(i);
44         }
45         reverse(all(revOut));
46         fill(all(vis), false);
47         for (auto node: revOut) {
48             if (vis[node]) continue;
49             comps.pb({});
50             dfs2(node);
51         }
52     }
53
54     vector<vector<int>> generateCondensedGraph() {
55         vector<vector<int>> adjCon(comps.size());
56         for (auto edge: edges)
57             if (compOf[edge.st] != compOf[edge.nd])
58                 adjCon[compOf[edge.st]].pb(compOf[edge.nd]);
59         return adjCon;
60     }
61 };

```

3.4 Satisfiability

3.4.1 2SAT

```

1 struct TwoSat {
2     int N;
3     vector<pair<int, int>> edges;
4
5     TwoSat(int _N = 0) {
6         init(_N);
7     }
8
9     void init(int _N = 0) {
10         N = _N;
11     }
12
13     int addVar() { return N++; }
14
15     /// x or y, edges will be refined in the end
16     void either(int x, int y) {
17         x = max(2 * x, -1 - 2 * x);
18         y = max(2 * y, -1 - 2 * y);
19         edges.pb({x, y});
20     }
21
22     void implies(int x, int y) {
23         either(-x, y);
24     }
25
26     void must(int x) {
27         either(x, x);
28     }
29
30     void XOR(int x, int y) {
31         either(x, y);
32         either(-x, -y);
33     }
34
35     vector<bool> solve(int _N = -1) {
36         if (_N != -1) N = _N;
37         SCC scc;
38         scc.init(2 * N);
39         for (auto e: edges) {
40             scc.addEdge(e.st ^ 1, e.nd);
41             scc.addEdge(e.nd ^ 1, e.st);
42         }
43         scc.gen();
44         for (int i = 0; i < 2 * N; ++i) {
45             if (scc.compOf[i] == scc.compOf[i ^ 1]) return {};
46         }
47         vector<vector<int>> comps = scc.comps;
48         reverse(all(comps));
49         vector<int> compOf(2 * N);
50         for (int i = 0; i < comps.size(); ++i) {
51             for (auto e: comps[i])
52                 compOf[e] = i;
53         }
54         vector<int> tmp(comps.size());
55         for (int i = 0; i < comps.size(); ++i) {
56             if (!tmp[i]) {
57                 tmp[i] = 1;
58                 for (auto e: comps[i])

```

```

59         tmp[compOf[e ^ 1]] = -1;
60     }
61 }
62 vector<bool> ans(N);
63 for (int i = 0; i < N; ++i)
64     ans[i] = tmp[compOf[2 * i]] == 1;
65 return ans;
66 }
67
68 void atMostOne(const vector<int> &li) {
69     if (li.size() <= 1) return;
70     int last = ~li[1];
71     for (int i = 2; i < li.size(); i++) {
72         int next = addVar();
73         implies(li[i], last);
74         either(last, next);
75         implies(li[i], next);
76         last = ~next;
77     }
78     implies(li[0], last);
79 }
80 };

```

3.5 Eulerian Path

3.5.1 EulerianPath

```

1 struct EulerianPath {
2     int n;
3     vector<vector<pair<int, int>>> adj; // {neighbor, edge_index}
4     vector<bool> used_edge;
5     vector<int> in_deg, out_deg;
6     vector<int> path;
7
8     EulerianPath(int n, int m) : n(n), adj(n + 1), used_edge(m, false),
9         in_deg(n + 1, 0), out_deg(n + 1, 0) {}
10
11     void add_edge(int u, int v, int edge_idx, bool directed = true) {
12         adj[u].push_back({v, edge_idx});
13         out_deg[u]++; in_deg[v]++;
14         if (!directed) {
15             adj[v].push_back({u, edge_idx});
16             out_deg[v]++; in_deg[u]++;
17         }
18
19     bool solve(int start_node) {
20         vector<int> ptr(n + 1, 0);
21         stack<int> st;
22         st.push(start_node);
23
24         while (!st.empty()) {
25             int u = st.top();
26             if (ptr[u] < adj[u].size()) {
27                 auto [v, id] = adj[u][ptr[u]++];
28                 if (!used_edge[id]) {
29                     used_edge[id] = true;
30                     st.push(v);
31                 }
32             } else {
33                 path.push_back(u);
34                 st.pop();
35             }
36         }
37         reverse(path.begin(), path.end());
38
39         // Check if all edges were visited
40         for (bool used : used_edge) {
41             if (!used) return false;
42         }
43         return true;
44     }
45 };

```

3.6 Flow and Min Cut

3.6.1 DinicMaxFlow

```

1 struct Dinic {
2     struct Edge {
3         int to, rev;
4         ll c, oc;
5         ll flow() { return max(oc - c, 0LL); } // if you need flows
6     };
7
8     vector<int> lvl, ptr, q;
9     vector<vector<Edge>> > adj;
10
11     Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {}
12
13     void addEdge(int a, int b, ll c, ll rcap = 0) {
14         adj[a].push_back({b, (int) adj[b].size(), c, c});
15         adj[b].push_back({a, (int) adj[a].size() - 1, rcap, rcap});
16     }
17
18     ll dfs(int v, int t, ll f) {
19         if (v == t || !f) return f;
20         for (int &i = ptr[v]; i < (int) adj[v].size(); i++) {
21             Edge &e = adj[v][i];
22             if (lvl[e.to] == lvl[v] + 1)
23                 if (ll p = dfs(e.to, t, min(f, e.c))) {
24                     e.c -= p, adj[e.to][e.rev].c += p;
25                     return p;
26                 }
27         }
28         return 0;
29     }
30
31     ll calc(int s, int t) {
32         ll flow = 0;
33         q[0] = s;
34         for (int L = 0; L < 31; L++)
35             do {

```

```

37         // 'int L=30' maybe faster for random data
38         lvl = ptr = vector<int>((int) q.size());
39         int qi = 0, qe = lvl[s] = 1;
40         while (qi < qe && !lvl[t]) {
41             int v = q[qi++];
42             for (Edge e : adj[v])
43                 if (!lvl[e.to] && e.c >> (30 - L))
44                     q[qi++] = e.to, lvl[e.to] = lvl[v] + 1;
45         }
46         while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
47         while (lvl[t]);
48         return flow;
49     }
50
51     bool leftOfMinCut(int a) { return lvl[a] != 0; }
52 };

```

3.6.2 MinCostMaxFlow

```

1 struct Edge {
2     int to;
3     int cost;
4     int cap, flow, backEdge;
5 };
6
7 struct MCMF
8 {
9
10     const int inf = 1000000010;
11     int n;
12     vector<vector<Edge>> g;
13
14     MCMF(int _n) {
15         n = _n + 1;
16         g.resize(n);
17     }
18
19     void addEdge(int u, int v, int cap, int cost) {
20         Edge e1 = {v, cost, cap, 0, (int) g[v].size()};
21         Edge e2 = {u, -cost, 0, 0, (int) g[u].size()};
22         g[u].push_back(e1);
23         g[v].push_back(e2);
24     }
25
26     pair<int, int> minCostMaxFlow(int s, int t) {
27         int flow = 0;
28         int cost = 0;
29         vector<int> state(n), from(n), from_edge(n);
30         vector<int> d(n);
31         deque<int> q;
32         while (true) {
33             for (int i = 0; i < n; i++)
34                 state[i] = 2, d[i] = inf, from[i] = -1;
35             state[s] = 1;
36             q.clear();
37             q.push_back(s);
38             d[s] = 0;
39             while (!q.empty()) {
40                 int v = q.front();
41                 q.pop_front();
42                 state[v] = 0;
43                 for (int i = 0; i < (int) g[v].size(); i++) {
44                     Edge e = g[v][i];
45                     if (e.flow >= e.cap || (d[e.to] <= d[v] + e.cost))
46                         continue;
47                     int to = e.to;
48                     d[to] = d[v] + e.cost;
49                     from[to] = v;
50                     from_edge[to] = i;
51                     if (state[to] == 1) continue;
52                     if (!state[to] || (!q.empty() && d[q.front()] > d[to]))
53                         q.push_front(to);
54                     else q.push_back(to);
55                     state[to] = 1;
56                 }
57             }
58             if (d[t] == inf) break;
59             int it = t, addflow = inf;
60             while (it != s) {
61                 addflow = min(addflow,
62                     g[from[it]][from_edge[it]].cap
63                     - g[from[it]][from_edge[it]].flow);
64                 it = from[it];
65             }
66             it = t;
67             while (it != s) {
68                 g[from[it]][from_edge[it]].flow += addflow;
69                 g[it][g[from[it]][from_edge[it]].backEdge].flow -=
70                     addflow;
71                 cost += g[from[it]][from_edge[it]].cost * addflow;
72                 it = from[it];
73             }
74             flow += addflow;
75             return {cost, flow};
76         }
77     };

```

3.7 Matching

3.7.1 HopcroftKarpMatching

```

1 struct HopcroftKarp {
2     vector<int> leftMatch, rightMatch, dist, cur;
3     vector<vector<int>> > a;
4     int n, m;
5
6     HopcroftKarp() {}
7
8     HopcroftKarp(int n, int m) {
9         this->n = n;
10        this->m = m;
11        a = vector<vector<int>> >(n);

```

```

12     leftMatch = vector<int>(m, -1);
13     rightMatch = vector<int>(n, -1);
14     dist = vector<int>(n, -1);
15     cur = vector<int>(n, -1);
16 }
17
18 void addEdge(int x, int y) {
19     a[x].push_back(y);
20 }
21
22 int bfs() {
23     int found = 0;
24     queue<int> q;
25     for (int i = 0; i < n; i++)
26         if (rightMatch[i] < 0) dist[i] = 0, q.push(i);
27         else dist[i] = -1;
28
29     while (!q.empty()) {
30         int x = q.front();
31         q.pop();
32         for (int i = 0; i < int(a[x].size()); i++) {
33             int y = a[x][i];
34             if (leftMatch[y] < 0) found = 1;
35             else if (dist[leftMatch[y]] < 0)
36                 dist[leftMatch[y]] = dist[x] + 1, q.push(leftMatch[
37                     y]);
38         }
39     }
40     return found;
41 }
42
43 int dfs(int x) {
44     for (; cur[x] < int(a[x].size()); cur[x]++) {
45         int y = a[x][cur[x]];
46         if (leftMatch[y] < 0 || (dist[leftMatch[y]] == dist[x] + 1
47             && dfs(leftMatch[y]))) {
48             leftMatch[y] = x;
49             rightMatch[x] = y;
50             return 1;
51         }
52     }
53     return 0;
54 }
55
56 int maxMatching() {
57     int match = 0;
58     while (bfs()) {
59         for (int i = 0; i < n; i++) cur[i] = 0;
60         for (int i = 0; i < n; i++)
61             if (rightMatch[i] < 0) match += dfs(i);
62     }
63     return match;
64 };

```

4 Number Theory

4.1 Euclidean Algorithm

4.1.1 GCD_LCM

```

1 ll gcd(ll a, ll b) {
2     if (b == 0)
3         return a;
4     return gcd(b, a % b);
5     // return __gcd(a, b);
6 }
7
8 ll lcm(ll a, ll b) {
9     return (a / gcd(a, b)) * b;
10 }

```

4.1.2 ExtendedEuclid

```

1 long long extGCD(long long a, long long b, long long &x, long long &y)
2 {
3     if (b == 0) {
4         x = 1; y = 0;
5         return a;
6     }
7     long long x1, y1;
8     long long d = extGCD(b, a % b, x1, y1);
9     x = y1;
10    y = x1 - y1 * (a / b);
11    return d;
12 }
13
14 long long modInverse(long long a, long long m) {
15     long long x, y;
16     long long g = extGCD(a, m, x, y);
17     if (g != 1) return -1; // Inverse doesn't exist
18     return (x % m + m) % m;
19 }

```

4.2 Primes and Divisors

4.2.1 SieveOfEratosthenes

```

1 bool isPrime[N];
2
3 void seive(int n = N) {
4     //complexity: n*log(log(n))
5     memset(isPrime, true, sizeof isPrime);
6     isPrime[0] = false;
7     isPrime[1] = false;
8     for (int i = 2; i * i < n; i++) {
9         if (isPrime[i]) {
10            for (int j = i * i; j < n; j += i) {
11                isPrime[j] = false;
12            }
13        }
14    }

```

```

15 }

```

4.2.2 SmallestPrimeFactor

```

1 vector<int> spf;
2
3 void seive(int n = N) {
4     //complexity: n*log(log(n))
5     spf.resize(n);
6     iota(all(spf), 0);
7     for (int i = 2; i * i < n; i++) {
8         if (spf[i] == i) {
9             for (int j = i * i; j < n; j += i) {
10                 spf[j] = min(spf[j], i);
11             }
12         }
13     }
14 }
15
16 vector<int> getPrimeFactors(int n) {
17     //complexity: log(n)
18     vector<int> ans;
19     while (n != 1) {
20         ans.pb(spf[n]);
21         n /= spf[n];
22     }
23     return ans;
24 }

```

4.2.3 DivisorsGeneration

```

1 template <typename T>
2 vector<T> getDivisors(T n) {
3     //complexity: sqrt(n)
4     vector<T> vec;
5     for (ll i = 1; i * i <= n; i++) {
6         if (n % i == 0) {
7             vec.push_back(i);
8             if (n / i != i)
9                 vec.push_back(n / i);
10        }
11    }
12    return vec;
13 }

```

4.2.4 PrimeFactorization

```

1 template <typename T>
2 vector<T> getPrimeFactors(T n) {
3     //complexity: sqrt(n)
4     vector<T> vec;
5     for (ll i = 2; i * i <= n; i++) {
6         while (n % i == 0) {
7             vec.push_back(i);
8             n /= i;
9         }
10    }
11    if (n != 1) vec.push_back(n);
12    return vec;
13 }

```

4.3 Modular Arithmetic

4.3.1 CRT

```

1 long long extGCD(long long a, long long b, long long &x, long long &y)
2 {
3     if (b == 0) {
4         x = 1; y = 0;
5         return a;
6     }
7     long long x1, y1;
8     long long d = extGCD(b, a % b, x1, y1);
9     x = y1;
10    y = x1 - y1 * (a / b);
11    return d;
12 }
13
14 // Solves x = a_i (mod m_i). Returns {x, lcm(m_1, ..., m_k)}.
15 pair<long long, long long> CRT(const vector<long long>& a, const vector
16     <long long>& m) {
17     long long res = a[0], lcm = m[0];
18     for (int i = 1; i < a.size(); i++) {
19         long long x, y;
20         long long g = extGCD(lcm, m[i], x, y);
21         if ((a[i] - res) % g != 0) return {-1, -1}; // No solution
22
23         long long step = lcm / g;
24         long long k = ((a[i] - res) / g) % step * (x % step) % step;
25         res += k * lcm;
26         lcm *= step;
27         res = (res % lcm + lcm) % lcm;
28     }
29     return {res, lcm};
30 }

```

5 Combinatorics

6 Math

6.1 Modular Arithmetic

6.1.1 ModularArithmetic

```

1 int fact[N], inv[N], invFact[N]; //used in nCr(), nPr(), calcFact()
2
3 int modSum(ll a, ll b) {
4     if (a < 0)
5         a += MOD;
6     if (b < 0)
7         b += MOD;
8     a += b;

```



```

9     if (a >= MOD)
10         a -= MOD;
11     return a;
12
13     a = (a % MOD + MOD) % MOD;
14     b = (b % MOD + MOD) % MOD;
15     return (a + b) % MOD;
16 }
17
18 int modProd(ll a, ll b) {
19     if (a < 0)
20         a += MOD;
21     if (b < 0)
22         b += MOD;
23     a *= b;
24     if (a >= MOD)
25         a %= MOD;
26     return a;
27
28     a = (a % MOD + MOD) % MOD;
29     b = (b % MOD + MOD) % MOD;
30     return (a * b) % MOD;
31 }
32
33
34 int modPow(int a, ll b) {
35     int result = 1;
36     while (b) {
37         if (b & 1)
38             result = modProd(result, a);
39         a = modProd(a, a);
40         b >>= 1;
41     }
42     return result;
43 }
44
45
46 int modInverse(int a) {
47     return modPow(a, MOD - 2);
48 }
49
50
51 int modDiv(int a, int b) {
52     return modProd(a, modInverse(b));
53 }
54
55 int nCr(int n, int r) {
56     if (r < 0 || r > n)
57         return 0;
58     // return modDiv(fact[n], modProd(fact[r], fact[n - r]));
59     return modProd(fact[n], modProd(invFact[r], invFact[n - r]));
60 }
61
62 int nPr(int n, int r) {
63     if (r < 0 || r > n)
64         return 0;
65     // return modDiv(fact[n], fact[n - r]);
66     return modProd(fact[n], invFact[n - r]);
67 }
68
69 void build() {
70     fact[0] = invFact[0] = 1;
71     for (int i = 1; i < N; i++) {
72         fact[i] = modProd(fact[i - 1], i);
73         invFact[i] = modDiv(invFact[i - 1], i);
74         inv[i] = modInverse(i);
75     }
76 }

```

6.1.2 ModularIntClass

```

1 struct Mint {
2     int v;
3
4     Mint(long long val = 0) {
5         v = int(val % MOD);
6         if (v < 0) v += MOD;
7     }
8
9     Mint operator+(const Mint& o) const { return Mint(v + o.v); }
10    Mint operator-(const Mint& o) const { return Mint(v - o.v); }
11    Mint operator*(const Mint& o) const { return Mint(1LL * v * o.v); }
12    Mint operator/(const Mint& o) const { return *this * o.inv(); }
13
14    Mint& operator+=(const Mint& o) {
15        v += o.v;
16        if (v >= MOD) v -= MOD;
17        return *this;
18    }
19
20    Mint& operator-=(const Mint& o) {
21        v -= o.v;
22        if (v < 0) v += MOD;
23        return *this;
24    }
25
26    Mint& operator*=(const Mint& o) {
27        v = int(1LL * v * o.v % MOD);
28        return *this;
29    }
30
31    Mint& operator/=(const Mint& o) {
32        v = int(1LL * v * o.inv() % MOD);
33        return *this;
34    }
35
36    Mint pow(long long p) const {
37        Mint a = *this, res = 1;
38        while (p > 0) {
39            if (p & 1) res *= a;
40            a *= a;
41            p >>= 1;
42        }
43        return res;

```

```

44    }
45
46    Mint inv() const { return pow(MOD - 2); }
47
48    friend ostream& operator<<(ostream& os, const Mint& m) {
49        os << m.v;
50        return os;
51    }
52 };
53
54 Mint fact[N], inv[N], invFact[N];
55
56 Mint nCr(int n, int r) {
57     if (r < 0 || r > n)
58         return 0;
59     // return fact[n] / (fact[r] * fact[n - r]);
60     return fact[n] * invFact[r] * invFact[n - r];
61 }
62
63 Mint nPr(int n, int r) {
64     if (r < 0 || r > n)
65         return 0;
66     // return fact[n] / fact[n - r];
67     return fact[n] * invFact[n - r];
68 }
69
70 void build() {
71     fact[0] = invFact[0] = 1;
72     for (int i = 1; i < N; i++) {
73         fact[i] = fact[i - 1] * i;
74         invFact[i] = invFact[i - 1] / i;
75         inv[i] = Mint(i).inv();
76     }
77 }

```

6.2 Multiplicative Functions

6.2.1 EulerTotientPhi

```

1 int phi[N];
2
3 void calculatePhi() {
4     for (int i = 0; i < N; ++i) phi[i] = i & 1 ? i : i / 2;
5     for (int i = 3; i < N; i += 2)
6         if (phi[i] == i)
7             for (int j = i; j < N; j += i) phi[j] -= phi[j] / i;
8 }

```

6.3 Matrices

6.3.1 MatrixExponentiation

```

1 int modSum(ll a, ll b) {
2     if (a < 0)
3         a += MOD;
4     if (b < 0)
5         b += MOD;
6     a += b;
7     if (a >= MOD)
8         a -= MOD;
9     return a;
10 }
11
12 int modProd(ll a, ll b) {
13     if (a < 0)
14         a += MOD;
15     if (b < 0)
16         b += MOD;
17     a *= b;
18     if (a >= MOD)
19         a %= MOD;
20     return a;
21 }
22
23 typedef vector<vector<int>> matrix;
24
25 matrix operator*(const matrix& lhs, const matrix& rhs) {
26     int n = lhs.size();
27     int m = rhs[0].size();
28     int s1 = lhs[0].size(), s2 = rhs.size();
29     assert(s1 == s2);
30     matrix ret(n, vector<int>(m));
31     for (int i = 0; i < n; ++i)
32         for (int j = 0; j < s1; ++j)
33             for (int k = 0; k < m; ++k)
34                 ret[i][k] = modSum(ret[i][k], modProd(lhs[i][j], rhs[j][k]));
35
36     return ret;
37 }
38
39 matrix Identity(int n) {
40     matrix ret(n, vector<int>(n));
41     for (int i = 0; i < n; ++i) {
42         ret[i][i] = 1;
43     }
44     return ret;
45 }
46
47 matrix mat_power(matrix x, ll p) {
48     matrix res = Identity(x.size());
49     while (p) {
50         if (p & 1) res = (res * x);
51         x = (x * x);
52         p >>= 1;
53     }
54     return res;

```

6.3.2 FastMatrixPower

```

1 const int SZ = 2, mod = 1e9 + 7;
2
3 int modSum(ll a, ll b) {

```



```

4     if (a < 0)
5         a += mod;
6     if (b < 0)
7         b += mod;
8     a += b;
9     if (a >= mod)
10        a -= mod;
11    return a;
12 }
13
14 int modProd(ll a, ll b) {
15     if (a < 0)
16         a += mod;
17     if (b < 0)
18         b += mod;
19     a *= b;
20     if (a >= mod)
21         a %= mod;
22     return a;
23 }
24
25 typedef array<array<int, SZ>, SZ> matrix;
26
27 matrix operator*(const matrix &lhs, const matrix &rhs) {
28     matrix ret{};
29     for (int i = 0; i < SZ; ++i)
30         for (int j = 0; j < SZ; ++j)
31             for (int k = 0; k < SZ; ++k)
32                 ret[i][k] = modSum(ret[i][k], modProd(lhs[i][j], rhs[j][k]));
33     return ret;
34 }
35
36 matrix Identity(int n) {
37     matrix ret = {};
38     for (int i = 0; i < n; ++i) {
39         ret[i][i] = 1;
40     }
41     return ret;
42 }
43
44 matrix mat_power(matrix x, ll p) {
45     matrix res = Identity(x.size());
46     while (p) {
47         if (p & 1) res = (res * x);
48         x = (x * x);
49         p >>= 1;
50     }
51     return res;
52 }

```

6.4 Linear Algebra

6.4.1 LinearXorBasis

```

1
2
3 struct Basis {
4     vector<ll> basis;
5     int LOG, sz;
6
7     Basis(int log = 60) {
8         LOG = log;
9         sz = 0;
10        basis.resize(LOG);
11    }
12
13    bool insert(ll x) {
14        for (int i = LOG - 1; i >= 0; --i) {
15            if (x >> i & 1 ^ 1) continue;
16            if (basis[i]) {
17                x ^= basis[i];
18            } else {
19                basis[i] = x;
20                sz++;
21                return true;
22            }
23        }
24        return false;
25    }
26
27    ll getMax(ll x = 0) {
28        ll ret = x;
29        for (int i = LOG - 1; i >= 0; --i) {
30            ret = max(ret, ret ^ basis[i]);
31        }
32        return ret;
33    }
34 };

```

6.5 Convolution

6.5.1 FastFourierTransform

```

1 #define rep(aa, bb, cc) for(int aa = bb; aa < cc; aa++)
2 #define sz(a) (int)a.size()
3 typedef complex<double> C;
4 typedef vector<double> vd;
5 void fft(vector<C>& a) {
6     int n = sz(a), L = 31 - __builtin_clz(n);
7     static vector<complex<long double>> R(2, 1);
8     static vector<C> rt(2, 1); // (~ 10% faster if double)
9     for (static int k = 2; k < n; k *= 2) {
10         R.resize(n); rt.resize(n);
11         auto x = polar(1.0L, acos(-1.0L) / k);
12         rep(i, k, 2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
13     }
14     vi rev(n);
15     rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
16     rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
17     for (int k = 1; k < n; k *= 2)
18         for (int i = 0; i < n; i += 2 * k) rep(j, 0, k) {
19             // C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-
20             // rolled) /// include-line

```

```

20         auto x = (double *)&rt[j+k], y = (double *)&a[i+j+k];
21             // exclude-line
22             C z(x[0]*y[0] - x[1]*y[1], x[0]*y[1] + x[1]*y[0]);
23             // exclude-line
24             a[i + j + k] = a[i + j] - z;
25             a[i + j] += z;
26         }
27     }
28 template<int M> vi convMod(const vi &a, const vi &b) {
29     if (a.empty() || b.empty()) return {};
30     vi res(sz(a) + sz(b) - 1);
31     int B=32-__builtin_clz(sz(res)), n=1<B, cut=int(sqrt(M));
32     vector<C> L(n), R(n), outs(n), outl(n);
33     rep(i, 0, sz(a)) L[i] = C((int)a[i] / cut, (int)a[i] % cut);
34     rep(i, 0, sz(b)) R[i] = C((int)b[i] / cut, (int)b[i] % cut);
35     fft(L), fft(R);
36     rep(i, 0, n) {
37         int j = -i & (n - 1);
38         outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
39         outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / 1i;
40     }
41     fft(outl), fft(outs);
42     rep(i, 0, sz(res)) {
43         ll av = 1l(real(outl[i])+.5), cv = 1l(imag(outs[i])+.5);
44         ll bv = 1l(imag(outl[i])+.5) + 1l(real(outs[i])+.5);
45         res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
46     }
47     return res;

```

6.5.2 NumberTheoreticTransform

```

1 const int MOD = 998244353;
2 const int ROOT = 3; // Primitive root for 998244353
3
4 long long power(long long base, long long exp) {
5     long long res = 1;
6     base %= MOD;
7     while (exp > 0) {
8         if (exp % 2 == 1) res = (res * base) % MOD;
9         base = (base * base) % MOD;
10        exp /= 2;
11    }
12    return res;
13 }
14
15 long long modInverseFast(long long n) {
16     return power(n, MOD - 2);
17 }
18
19 void ntt(vector<long long>& a, bool invert) {
20     int n = a.size();
21     for (int i = 1, j = 0; i < n; i++) {
22         int bit = n >> 1;
23         for (; j & bit; bit >>= 1) j ^= bit;
24         j ^= bit;
25         if (i < j) swap(a[i], a[j]);
26     }
27     for (int len = 2; len <= n; len <= 1) {
28         long long wlen = power(ROOT, (MOD - 1) / len);
29         if (invert) wlen = modInverseFast(wlen);
30         for (int i = 0; i < n; i += len) {
31             long long w = 1;
32             for (int j = 0; j < len / 2; j++) {
33                 long long u = a[i + j], v = (a[i + j + len / 2] * w) % MOD;
34                 a[i + j] = u + v < MOD ? u + v : u + v - MOD;
35                 a[i + j + len / 2] = u - v >= 0 ? u - v : u - v + MOD;
36                 w = (w * wlen) % MOD;
37             }
38         }
39     }
40     if (invert) {
41         long long n_inv = modInverseFast(n);
42         for (long long& x : a) x = (x * n_inv) % MOD;
43     }
44 }

```

7 Strings

7.1 String Matching

7.1.1 KMP_PrefixFunction

```

1 vector<int> compute_pi(const string &s) {
2     int n = s.size();
3     vector<int> pi(n);
4     for (int i = 1; i < n; i++) {
5         int j = pi[i - 1];
6         while (j > 0 && s[i] != s[j])
7             j = pi[j - 1];
8         if (s[i] == s[j])
9             j++;
10        pi[i] = j;
11    }
12    return pi;
13 }
14
15 vector<int> find_matches(const string &text, const string &pat) {
16     int n = pat.length(), m = text.length();
17     string s = pat + "#" + text;
18     vector<int> pi = compute_pi(s), ans;
19     for (int i = n + 1; i <= n + m; i++) { /* n + 1 is where the text
20         starts */
21         if (pi[i] == n) {
22             ans.push_back(i - 2 * n); /* i - (n - 1) - (n + 1) */
23         }
24     }
25     return ans;
26 }
27 void compute_automaton(string s, vector<vector<int>> &aut) {

```

```

28     s += '#';
29     int n = s.size();
30     vector<int> pi = compute_pi(s);
31     aut.assign(n, vector<int>(26));
32     for (int i = 0; i < n; i++) {
33         for (int c = 0; c < 26; c++) {
34             if (i > 0 && 'a' + c != s[i])
35                 aut[i][c] = aut[pi[i - 1]][c];
36             else
37                 aut[i][c] = i + ('a' + c == s[i]);
38         }
39     }
40 }

```

7.1.2 RabinKarp

```

1 vector<int> rabin_karp(string const &s, string const &t) {
2     const int p = 31;
3     const int m = 1e9 + 9;
4     int S = s.size(), T = t.size();
5
6     vector<ll> p_pow(max(S, T));
7     p_pow[0] = 1;
8     for (int i = 1; i < (int) p_pow.size(); i++)
9         p_pow[i] = (p_pow[i - 1] * p) % m;
10
11     vector<ll> h(T + 1, 0);
12     for (int i = 0; i < T; i++)
13         h[i + 1] = (h[i] + (t[i] - 'a' + 1) * p_pow[i]) % m;
14     ll h_s = 0;
15     for (int i = 0; i < S; i++)
16         h_s = (h_s + (s[i] - 'a' + 1) * p_pow[i]) % m;
17
18     vector<int> occurrences;
19     for (int i = 0; i + S - 1 < T; i++) {
20         ll cur_h = (h[i + S] + m - h[i]) % m;
21         if (cur_h == h_s * p_pow[i] % m)
22             occurrences.push_back(i);
23     }
24     return occurrences;
25 }

```

7.1.3 ZFunction

```

1 // z[i] is the length of the longest substring starting from s[i] which
2 // is also a prefix of s
3 vector<int> z_function(string s) {
4     int n = s.length();
5     vector<int> z(n);
6     int l = 0, r = 0;
7     for (int i = 1; i < n; i++) {
8         if (i < r) {
9             z[i] = min(r - i, z[i - l]);
10        }
11        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
12            z[i]++;
13        }
14        if (i + z[i] > r) {
15            l = i;
16            r = i + z[i];
17        }
18    }
19    return z;
20 }

```

7.2 Hashing

7.2.1 DoubleStringHashing

```

1 int modSum(ll a, ll b) {
2     ///complexity: 1
3     if (a < 0)
4         a += MOD;
5     if (b < 0)
6         b += MOD;
7     a += b;
8     if (a >= MOD)
9         a %= MOD;
10    return a;
11 }
12
13 int modProd(ll a, ll b) {
14     ///complexity: 1
15     if (a < 0)
16         a += MOD;
17     if (b < 0)
18         b += MOD;
19     a *= b;
20     if (a >= MOD)
21         a %= MOD;
22    return a;
23 }
24
25 int fastPow(int a, ll b) {
26     ///complexity: log(b)
27     if (b == 0) return 1;
28     int temp = fastPow(a, b / 2);
29     if (b % 2 == 0)
30         return modProd(temp, temp);
31     return modProd(modProd(a, temp), temp);
32 }
33
34
35 int modInverse(int a) {
36     ///complexity: log(mod)
37     return fastPow(a, MOD - 2);
38 }
39
40
41 struct Hash {
42     vector<pair<int, int>> hash_value = {{0, 0}};
43     const pair<int, int> p = {67, 97};
44     vector<pair<int, int>> p_pow = {{1, 1}};

```

```

45     vector<pair<int, int>> inv = {{1, 1}};
46
47     void compute(string const &s) {
48         for (char c: s) {
49             if (c >= 'a' && c <= 'z') {
50                 c = c - 'a' + 1;
51             } else if (c >= 'A' && c <= 'Z') {
52                 c = c - 'A' + 30;
53             } else {
54                 c = c - '0' + 60;
55             }
56             int x = modSum(hash_value.back().st, modProd(c, p_pow.back().st));
57             int a = modProd(p_pow.back().st, p.st);
58
59             int y = modSum(hash_value.back().nd, modProd(c, p_pow.back().nd));
60             int b = modProd(p_pow.back().nd, p.nd);
61
62             hash_value.emplace_back(x, y);
63             p_pow.emplace_back(a, b);
64             if (inv.size() == 1) {
65                 inv.emplace_back(modInverse(a), modInverse(b));
66             } else {
67                 inv.emplace_back(modProd(inv[1].st, inv.back().st),
68                                     modProd(inv[1].nd, inv.back().nd));
69             }
70         }
71
72         pair<int, int> get(int l, int r) {
73             l++, r++;
74             pair<int, int> ans;
75             ans.st = modSum(hash_value[r].st, -hash_value[l - 1].st);
76             ans.st = modProd(ans.st, inv[l].st);
77
78             ans.nd = modSum(hash_value[r].nd, -hash_value[l - 1].nd);
79             ans.nd = modProd(ans.nd, inv[l].nd);
80
81             return ans;
82         }
83     }
84 };

```

7.3 Aho Corasick

7.3.1 AhoCorasickAutomaton

```

1 struct Mapper {
2     int operator()(char c) const { return c - 'a'; }
3 };
4
5 template <int ALPHABET, typename Mapper>
6 struct Aho {
7     struct Node {
8         array<int, ALPHABET> nxt;
9         int link = 0;
10        int terminal_idx = -1; // -1 if not a terminal, else idx of string insertion
11        int exit_link = 0; // Points to the nearest terminal suffix
12        int depth = 0; // Size of the prefix the node represents
13        Node() { nxt.fill(0); }
14    };
15
16    vector<Node> t;
17    vector<int> bfs_order;
18    Mapper get_idx; // Instance of the mapping function
19    int word_count = 0; // Tracks the order of inserted strings
20
21    // Constructor optionally accepts a custom mapper instance
22    Aho(Mapper mapper = Mapper()) : get_idx(mapper) {
23        t.emplace_back();
24    }
25
26    int insert(const string& p) {
27        int u = 0;
28        for (char c : p) {
29            int v = get_idx(c);
30            if (!t[u].nxt[v]) {
31                t[u].nxt[v] = t.size();
32                t.emplace_back();
33                t.back().depth = t[u].depth + 1; // New node's depth is parent's depth + 1
34            }
35            u = t[u].nxt[v];
36        }
37
38        // If this exact string hasn't been inserted before, assign it the current index
39        if (t[u].terminal_idx == -1) {
40            t[u].terminal_idx = word_count;
41        }
42        word_count++; // Increment for the next insertion
43
44        return u;
45    }
46
47    void build() {
48        queue<int> q;
49        for (int c = 0; c < ALPHABET; ++c) {
50            if (t[0].nxt[c]) {
51                q.push(t[0].nxt[c]);
52            }
53        }
54
55        while (!q.empty()) {
56            int u = q.front();
57            q.pop();
58            bfs_order.push_back(u);
59
60            for (int c = 0; c < ALPHABET; ++c) {
61                int v = t[u].nxt[c];
62                if (v) {
63                    t[v].link = t[t[u].link].nxt[c];

```

```

64         // Compute the exit link:
65         // If the immediate suffix link is a terminal,
66         // point to it.
67         // Otherwise, copy its exit link.
68         t[v].exit_link = (t[t[v].link].terminal_idx != -1)
69             ? t[v].link : t[t[v].link].exit_link;
70
71         q.push(v);
72     }
73     else {
74         t[u].nxt[c] = t[t[u].link].nxt[c];
75     }
76 }
77
78 int advance(int u, char c) {
79     return t[u].nxt[get_idx(c)];
80 }
81
82 };

```

7.4 Suffix Structures

7.4.1 SuffixArrayLCP

```

1 struct SuffixArray {
2     string S;
3     // sa is the suffix array with the empty suffix being sa[0]
4     // lcp[i] holds the lcp between sa[i], sa[i - 1]
5     vector<int> logs, sa, lcp, rank;
6     vector<vector<int>> table;
7
8     SuffixArray() {};
9
10    SuffixArray(string &s, int lim = 256) {
11        S = s;
12        int n = s.size() + 1, k = 0, a, b;
13        vector<int> c(s.begin(), s.end() + 1), tmp(n), frq(max(n, lim))
14        ;
15        c.back() = 0; //0 is less than any character
16        sa = lcp = rank = tmp, iota(sa.begin(), sa.end(), 0);
17        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
18            p = j, iota(tmp.begin(), tmp.end(), n - j);
19            for (int i = 0; i < n; i++) {
20                if (sa[i] >= j)
21                    tmp[p++] = sa[i] - j;
22            }
23
24            fill(frq.begin(), frq.end(), 0);
25
26            for (int i = 0; i < n; i++) frq[c[i]]++;
27            for (int i = 1; i < lim; i++) frq[i] += frq[i - 1];
28            for (int i = n; i--;) sa[--frq[c[tmp[i]]]] = tmp[i];
29
30            swap(c, tmp), p = 1, c[sa[0]] = 0;
31            for (int i = 1; i < n; i++)
32                a = sa[i - 1], b = sa[i], c[b] = (tmp[a] == tmp[b] &&
33                    tmp[a + j] == tmp[b + j]) ? p - 1 : p++;
34        }
35
36        for (int i = 1; i < n; i++) rank[sa[i]] = i;
37        for (int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
38            for (k &&k--, j = sa[rank[i] - 1];
39                s[i + k] == s[j + k];
40                k++);
41
42        void preLcp() {
43            int n = S.size() + 1;
44            logs = vector<int>(n + 5);
45            for (int i = 2; i < n + 5; ++i) {
46                logs[i] = logs[i / 2] + 1;
47            }
48            table = vector<vector<int>>(n, vector<int>(20));
49            for (int i = 0; i < n; ++i) {
50                table[i][0] = lcp[i];
51            }
52
53            for (int j = 1; j <= logs[n]; ++j) {
54                for (int i = 0; i <= n - (1 << j); ++i) {
55                    table[i][j] = min(table[i][j - 1], table[i + (1 << (j -
56                        1))][j - 1]);
57                }
58            }
59
60            int queryLcp(int i, int j) {
61                // if (i == j) return (int) S.size() - i;
62                // i = rank[i], j = rank[j];
63                if (i == j) return (int) S.size() - sa[i];
64                if (i > j)
65                    swap(i, j);
66                i++;
67                int len = logs[j - i + 1];
68                return min(table[i][len], table[j - (1 << len) + 1][len]);
69            }
70        }
71    };

```

7.4.2 SuffixAutomaton

```

1 struct SuffixAutomaton {
2     struct Node {
3         int len = 0, link = -1;
4         map<char, int> next;
5     };
6
7     vector<Node> st;
8     int sz = 1, last = 0;
9
10    SuffixAutomaton(const string& s) {
11        st.resize(s.length() * 2);
12        st[0].len = 0;
13        st[0].link = -1;

```

```

14        for (char c : s) extend(c);
15    }
16
17    void extend(char c) {
18        int cur = sz++;
19        st[cur].len = st[last].len + 1;
20        int p = last;
21        while (p != -1 && !st[p].next.count(c)) {
22            st[p].next[c] = cur;
23            p = st[p].link;
24        }
25        if (p == -1) {
26            st[cur].link = 0;
27        } else {
28            int q = st[p].next[c];
29            if (st[p].len + 1 == st[q].len) {
30                st[cur].link = q;
31            } else {
32                int clone = sz++;
33                st[clone].len = st[p].len + 1;
34                st[clone].next = st[q].next;
35                st[clone].link = st[q].link;
36                while (p != -1 && st[p].next[c] == q) {
37                    st[p].next[c] = clone;
38                    p = st[p].link;
39                }
40                st[q].link = st[cur].link = clone;
41            }
42        }
43        last = cur;
44    }
45 };

```

7.5 Palindromes

7.5.1 Manacher

```

1 struct Manacher {
2     vector<int> p;
3
4     // Returns a vector where p[i] is the radius of the longest
5     // palindrome around center i
6     // Centers are interleaved: 0 is before string, 1 is first char, 2
7     // is between 1st & 2nd, etc.
8     Manacher(string s) {
9         string t = "#";
10        for (char c : s) {
11            t += c;
12            t += "#";
13        }
14        int n = t.size();
15        p.assign(n, 0);
16        int c = 0, r = 0;
17        for (int i = 0; i < n; i++) {
18            int mirror = 2 * c - i;
19            if (i < r) p[i] = min(r - i, p[mirror]);
20            while (i - 1 - p[i] >= 0 && i + 1 + p[i] < n && t[i - 1 - p
21                [i]] == t[i + 1 + p[i]]) {
22                p[i]++;
23            }
24            if (i + p[i] > r) {
25                c = i;
26                r = i + p[i];
27            }
28        }
29
30        // Check if substring [l, r] (0-based) is a palindrome in O(1)
31        bool is_palindrome(int l, int r) {
32            int len = r - l + 1;
33            int center = l + r + 1;
34            return p[center] >= len;
35        }
36    };

```

8 Dynamic Programming (DP)

8.1 DP Optimizations

8.1.1 DivideAndConquerDP

```

1 int a[N];
2 ll curCost;
3 int curL = 0, curR = -1, freq[N];
4 pair<int, ll> pre[N], cur[N];
5 // Modify at will :
6 int lo(int ind) { return 1; }
7 int hi(int ind) { return ind; }
8
9 void add(int val) {
10    curCost += freq[val]++;
11 }
12
13 void remove(int val) {
14    curCost -= --freq[val];
15 }
16
17 ll Cost(int l, int r) {
18    while (curR < r)
19        add(a[++curR]);
20    while (curL > l)
21        add(a[--curL]);
22    while (curR > r)
23        remove(a[curR--]);
24    while (curL < l)
25        remove(a[curL++]);
26    return curCost;
27 }
28
29 ll f(int ind, int k) { return pre[k - 1].nd + Cost(k, ind); }
30 void store(int ind, int k, ll v) { cur[ind] = pair(k, v); }
31
32 void rec(int L, int R, int LO, int HI) {

```

```

33     if (L > R) return;
34     int mid = (L + R) >> 1;
35     pair best(llinf, L0);
36     for (int k = max(L0, lo(mid)); k <= min(HI, hi(mid)); ++k)
37         best = min(best, pair(f(mid, k), k));
38     store(mid, best.second, best.first);
39     rec(L, mid - 1, L0, best.second);
40     rec(mid + 1, R, best.second, HI);
41 }
42
43 void solve(int L, int R) { rec(L, R, -inf, inf); }

```

8.1.2 ConvexHullTrick

```

1 //To query the maximum value of the function for all x in [l, r], query
  //only two points: l, r because the functions are convex.
2 struct Line {
3     ll m, b;
4     mutable function<const Line *(>> succ;
5
6     bool operator<(const Line &other) const {
7         return m < other.m;
8     }
9
10    bool operator<(const ll &x) const {
11        const Line *s = succ();
12        if (!s)
13            return 0;
14        return b - s->b < (s->m - m) * x;
15    }
16 };
17
18 // will maintain upper hull for maximum
19 struct HullDynamic : public multiset<Line, less<>> {
20     bool bad(iterator y) {
21         auto z = next(y);
22         if (y == begin()) {
23             if (z == end())
24                 return 0;
25             return y->m == z->m && y->b <= z->b;
26         }
27         auto x = prev(y);
28         if (z == end())
29             return y->m == x->m && y->b <= x->b;
30         return (ld)(x->b - y->b) * (z->m - y->m) >= (ld)(y->b - z->b)
31             * (y->m - x->m);
32     }
33
34     void insert_line(ll m, ll b) {
35         // for minimum
36         // m *= -1;
37         // b *= -1;
38         auto y = insert({m, b});
39         y->succ = [=] { return next(y) == end() ? 0 : &*next(y); };
40         if (bad(y)) {
41             erase(y);
42             return;
43         }
44         while (next(y) != end() && bad(next(y)))
45             erase(next(y));
46         while (y != begin() && bad(prev(y)))
47             erase(prev(y));
48     }
49
50     ll query(ll x) {
51         auto l = *lower_bound(x);
52         // for minimum
53         // return -(l.m * x + l.b);
54         return l.m * x + l.b;
55     }
56 };

```

8.1.3 ConvexHullTrickRollback

```

1 // Template parameters:
2 // IS_MAX: Set to true for Maximum queries, false for Minimum queries.
3 // IS_INC: Set to true if slopes added are strictly Increasing, false
  // for Decreasing.
4 template<bool IS_MAX = false, bool IS_INC = false>
5 struct RollbackCHT {
6     struct Line {
7         ll m, c;
8         Line() : m(0), c(llinf) {}
9         Line(ll m, ll c) : m(m), c(c) {}
10        ll eval(ll x) const { return m * x + c; }
11    };
12
13    struct History {
14        int pos;
15        int sz;
16        Line old_line;
17    };
18
19    vector<Line> st;
20    vector<History> hist;
21    int sz;
22
23    RollbackCHT(int max_nodes) {
24        st.resize(max_nodes);
25        sz = 0;
26    }
27
28    bool redundant(Line l1, Line l2, Line l3) {
29        __int128 dx1 = l1.m - l2.m;
30        __int128 dc1 = l2.c - l1.c;
31        __int128 dx2 = l2.m - l3.m;
32        __int128 dc2 = l3.c - l2.c;
33        return dc1 * dx2 >= dc2 * dx1;
34    }
35
36    void add(ll m, ll c) {
37        // The magic happens here: automatically formats internal state
38        ll m_int = IS_INC ? -m : m;
39        ll c_int = IS_MAX ? -c : c;

```

```

40        Line l_new(m_int, c_int);
41        int pos = sz;
42        int low = 1, high = sz - 1;
43
44        while (low <= high) {
45            int mid = low + (high - low) / 2;
46            if (redundant(st[mid - 1], st[mid], l_new)) {
47                pos = mid;
48                high = mid - 1;
49            } else {
50                low = mid + 1;
51            }
52        }
53
54        hist.push_back({pos, sz, st[pos]});
55        st[pos] = l_new;
56        sz = pos + 1;
57    }
58
59    void rollback() {
60        if (hist.empty()) return;
61        History h = hist.back();
62        hist.pop_back();
63        sz = h.sz;
64        st[h.pos] = h.old_line;
65    }
66
67    ll query(ll x) {
68        if (sz == 0) return IS_MAX ? -llinf : llinf;
69
70        // The magic happens here: matches x to the internal geometry
71        ll x_int = (IS_INC != IS_MAX) ? -x : x;
72
73        int low = 0, high = sz - 2;
74        int best = sz - 1;
75
76        while (low <= high) {
77            int mid = low + (high - low) / 2;
78            if (st[mid].eval(x_int) <= st[mid + 1].eval(x_int)) {
79                best = mid;
80                high = mid - 1;
81            } else {
82                low = mid + 1;
83            }
84        }
85
86        ll ans = st[best].eval(x_int);
87        return IS_MAX ? -ans : ans;
88    }
89
90 };

```

8.1.4 LiChaoTree

```

1 // To get an instance : node *root = EMPTY;
2 typedef pair<long long, long long> line;
3 int start_x, end_x;
4 line neutral(0, -1e18);
5
6 ll sub(const line &l, ll x) {
7     return l.first * x + l.second;
8 }
9
10 double inter(const line &l1, const line &l2) {
11     return (l2.second - l1.second) / (l1.first - l2.first - .0);
12 }
13
14 extern struct node *const EMPTY;
15 struct node {
16     line l;
17     node *lf, *rt;
18     node() : l(neutral), lf(this), rt(this) {}
19     node(line l) : l(l), lf(EMPTY), rt(EMPTY) {}
20 };
21 node *const EMPTY = new node;
22
23 void insert(line l, int lx, int rx, node *&cur, ll ns = start_x, ll ne = end_x) {
24     if (rx < ns || lx > ne) return;
25     if (ns >= lx && ne <= rx) {
26         if (cur == EMPTY) {
27             cur = new node(l);
28             return;
29         }
30         if (l.first == cur->l.first) {
31             cur->l = max(cur->l, l);
32             return;
33         }
34
35         auto x = inter(l, cur->l);
36         if (x < ns || x > ne) {
37             if (sub(l, ns) > sub(cur->l, ns)) cur->l = l;
38             return;
39         }
40         ll mid = ns + (ne - ns) / 2;
41         if (x < mid) {
42             if (sub(l, ne) > sub(cur->l, ne)) swap(l, cur->l);
43             insert(l, lx, rx, cur->lf, ns, mid);
44         } else {
45             if (sub(l, ns) > sub(cur->l, ns)) swap(l, cur->l);
46             insert(l, lx, rx, cur->rt, mid + 1, ne);
47         }
48         return;
49     }
50
51     if (cur == EMPTY) {
52         cur = new node(neutral);
53     }
54     ll mid = ns + (ne - ns) / 2;
55     insert(l, lx, rx, cur->lf, ns, mid);
56     insert(l, lx, rx, cur->rt, mid + 1, ne);
57 }
58
59 ll query(ll x, node *cur, ll ns = start_x, ll ne = end_x) {

```

```

60     auto ret = sub(cur->l, x);
61     if (x < ns || x > ne)
62         return sub(neutral, x);
63     if (ns == ne)
64         return ret;
65     ll mid = ns + (ne - ns) / 2;
66     if (x <= mid)
67         return max(ret, query(x, cur->lf, ns, mid));
68     return max(ret, query(x, cur->rt, mid + 1, ne));
69 }

```

8.1.5 PersistentLiChaoTree

```

1  const ll maxN = 1e7 + 5;
2
3  struct Line {
4      ll m, c;
5
6      Line() : m(0), c(1llinf) {}
7
8      Line(ll m, ll c) : m(m), c(c) {}
9  };
10
11 ll sub(ll x, Line l) {
12     return x * l.m + l.c;
13 }
14
15 // Persistent Li Chao
16 struct Node {
17     // range I am responsible for
18     Line line;
19     Node *left, *right;
20
21     Node() {
22         left = right = NULL;
23     }
24
25     Node(ll m, ll c) {
26         line = Line(m, c);
27         left = right = NULL;
28     }
29
30     void extend(int l, int r) {
31         if (left == NULL && l != r) {
32             left = new Node();
33             right = new Node();
34         }
35     }
36
37     Node *copy(Node *node) {
38         Node *newNode = new Node;
39         newNode->left = node->left;
40         newNode->right = node->right;
41         newNode->line = node->line;
42         return newNode;
43     }
44
45     Node *add(Line toAdd, int l, int r) {
46         int mid = (l + r) / 2;
47         Node *cur = copy(this);
48         if (l == r) {
49             if (sub(l, toAdd) < sub(l, cur->line))
50                 swap(toAdd, cur->line);
51             return cur;
52         }
53         bool lef = sub(l, toAdd) < sub(l, cur->line);
54         bool midE = sub(mid + 1, toAdd) < sub(mid + 1, cur->line);
55         if (midE)
56             swap(cur->line, toAdd);
57         cur->extend(l, r);
58         if (lef != midE) {
59             cur->left = cur->left->add(toAdd, l, mid);
60         } else {
61             if (mid + 1 > r)
62                 return cur;
63             cur->right = cur->right->add(toAdd, mid + 1, r);
64         }
65         return cur;
66     }
67
68     Node *add(Line toAdd) {
69         return add(toAdd, -maxN + 1, maxN - 1);
70     }
71
72     ll query(ll x, int l, int r) {
73         int mid = (l + r) / 2;
74         if (l == r || left == NULL)
75             return sub(x, line);
76         extend(l, r);
77         if (x <= mid)
78             return min(sub(x, line), left->query(x, l, mid));
79         else
80             return min(sub(x, line), right->query(x, mid + 1, r));
81     }
82
83     ll query(ll x) {
84         return query(x, -maxN + 1, maxN - 1);
85     }
86
87     void clear() {
88         if (left != NULL) {
89             left->clear();
90             right->clear();
91         }
92         delete this;
93     }
94 };
95
96 Node *tree[N];

```

8.2 Bitmask DP

8.2.1 SOS_DP

```

1 // Computes the sum of all subsets for every bitmask in O(N * 2^N)
2 vector<long long> SOS_DP(int n_bits, const vector<long long>& a) {
3     int max_mask = 1 << n_bits;
4     vector<long long> dp = a; // dp[mask] initially holds exactly the
5                               // value for mask
6
7     for (int i = 0; i < n_bits; ++i) {
8         for (int mask = 0; mask < max_mask; ++mask) {
9             if (mask & (1 << i)) {
10                 dp[mask] += dp[mask ^ (1 << i)];
11             }
12         }
13     }
14     return dp; // dp[mask] now contains sum of all submasks of mask

```

9 Trees

9.1 Lowest Common Ancestor (LCA)

9.1.1 BinaryLiftingLCA

```

1 struct LCA {
2     static const int B = 21;
3     vector<array<int, 2>> tour;
4     vector<array<array<int, 2>, B>> T;
5     vector<int> d, in, lg;
6
7     LCA(int n, auto &adj): d(n + 1), in(n + 1), lg(n*2 + 1) {
8         tour.reserve(n * 2);
9         T.resize(n);
10        dfs(0, 0, adj);
11        for(int i = 0; i < n; ++i) lg[i]=__lg(i), T[i][0]=tour[i];
12        for (int j = 1; j < B; ++j)
13            for (int i = 0; i + (1 << j) - 1 < n; ++i)
14                T[i][j] = min(T[i][j - 1], T[i+(1 << j - 1)][j - 1]);
15    }
16
17    void dfs(int u, int p, auto &adj) {
18        in[u] = size(tour);
19        tour.push_back({d[u], u});
20        for (int v: adj[u]) {
21            if (v == p) continue;
22            d[v] = d[u] + 1;
23            dfs(v, u, adj);
24            tour.push_back({d[u], u});
25        }
26    }
27
28    int get(int u, int v) {
29        int l = min(in[u], in[v]), r = max(in[u], in[v]);
30        int j = lg[r - l + 1];
31        return min(T[l][j], T[r - (1 << j) + 1][j])[1];
32    }
33
34    int dist(int u, int v) { return d[u]+d[v]-2*d[get(u, v)]; }
35 };

```

9.2 DSU on Tree

9.2.1 DSUOnTree_Sack

```

1 class DSUOnTree {
2 private:
3     struct Query {
4         int idx, x;
5     };
6
7     int n;
8     vector<vector<int>> adj;
9     vector<int> sz, depth;
10    vector<bool> big;
11    vector<vector<Query>> queries;
12    vector<int> ans;
13
14    // =====
15    // PROBLEM SPECIFIC STATE VARIABLES
16    // =====
17    vector<char> c; // Example: Array of characters per node
18
19    // Example state variables
20    // int distinct_count = 0;
21    // vector<int> freq;
22
23    void add(int u, int p) {
24        // --- ADD LOGIC HERE ---
25        // Example: freq[c[u] - 'a']++;
26
27        for (int v : adj[u]) {
28            if (v == p || big[v]) continue; // Skip parent and heavy
29                                           // child[cite: 4]
30            add(v, u);
31        }
32    }
33
34    void remove(int u, int p) {
35        // --- REMOVE LOGIC HERE ---
36        // Example: freq[c[u] - 'a']--;
37
38        for (int v : adj[u]) {
39            if (v == p || big[v]) continue; // Skip parent and heavy
40                                           // child[cite: 4]
41            remove(v, u);
42        }
43    }
44
45    int get_answer(int u, int x = 0) {
46        // --- QUERY ANSWER LOGIC HERE ---
47        return 0;
48    }
49    // =====
50
51    void pre(int u, int p = 0) {

```

```

50     sz[u] = 1;
51     depth[u] = depth[p] + 1;
52     for (int v : adj[u]) {
53         if (v == p) continue;
54         pre(v, u);
55         sz[u] += sz[v]; // Compute subtree size[cite: 4]
56     }
57 }
58
59 void dfs(int u, int p = 0, bool keep = false) {
60     int mx = -1, bigChild = -1;
61
62     // Find the heavy child[cite: 4]
63     for (int v : adj[u]) {
64         if (v == p) continue;
65         if (sz[v] > mx) {
66             mx = sz[v];
67             bigChild = v;
68         }
69     }
70
71     // Process light children and clear them[cite: 4]
72     for (int v : adj[u]) {
73         if (v == p || v == bigChild) continue;
74         dfs(v, u, false);
75     }
76
77     // Process heavy child and keep it[cite: 4]
78     if (bigChild != -1) {
79         dfs(bigChild, u, true);
80         big[bigChild] = true;
81     }
82
83     // Add current node and its light children[cite: 4]
84     add(u, p);
85
86     // Answer all queries for the current node[cite: 4]
87     for (const auto& q : queries[u]) {
88         ans[q.idx] = get_answer(u, q.x);
89     }
90
91     // Reset the big child flag[cite: 4]
92     if (bigChild != -1) {
93         big[bigChild] = false;
94     }
95
96     // Clear state if this node is not meant to be kept[cite: 4]
97     if (!keep) {
98         remove(u, p);
99     }
100 }
101
102 public:
103     // Initialize with 1-based indexing in mind
104     DSUOnTree(int nodes, const vector<char>& chars) : n(nodes), c(chars) {
105         adj.assign(n + 1, vector<int>());
106         sz.assign(n + 1, 0);
107         depth.assign(n + 1, 0);
108         big.assign(n + 1, false);
109         queries.assign(n + 1, vector<Query>());
110         // Initialize problem-specific arrays here (e.g., freq.assign(26, 0);)
111     }
112
113     void add_edge(int u, int v) {
114         adj[u].push_back(v);
115         adj[v].push_back(u);
116     }
117
118     void add_query(int node, int query_idx, int x = 0) {
119         queries[node].push_back({query_idx, x});
120     }
121
122     vector<int> solve(int root, int num_queries) {
123         ans.assign(num_queries, 0);
124         pre(root, 0); // Precalculate sizes and depths[cite: 4]
125         dfs(root, 0, false); // Run the sack algorithm[cite: 4]
126         return ans;
127     }
128 };

```

9.3 Heavy Light Decomposition (HLD)

9.3.1 HeavyLightDecomposition

```

1 class HLD {
2 public:
3     vector<int> par, sz, head, tin, tout, who, depth;
4
5     int dfs1(int u, vector<vector<int>> &adj) {
6         for (int &v: adj[u]) {
7             if (v == par[u]) continue;
8             depth[v] = depth[u] + 1;
9             par[v] = u;
10            sz[u] += dfs1(v, adj);
11            if (sz[v] > sz[adj[u][0]] || adj[u][0] == par[u]) swap(v, adj[u][0]);
12        }
13        return sz[u];
14    }
15
16    void dfs2(int u, int &timer, const vector<vector<int>> &adj) {
17        tin[u] = timer++;
18        for (int v: adj[u]) {
19            if (v == par[u]) continue;
20            head[v] = (timer == tin[u] + 1 ? head[u] : v);
21            dfs2(v, timer, adj);
22        }
23        tout[u] = timer - 1;
24    }
25
26    HLD(vector<vector<int>> &adj, int r = 0)

```

```

27        : par(adj.size(), -1), sz(adj.size(), 1), head(adj.size(), r),
28            tin(adj.size()), tout(adj.size()),
29            who(adj.size()),
30            depth(adj.size()) {
31            dfs1(r, adj);
32            int x = 0;
33            dfs2(r, x, adj);
34            for (int i = 0; i < adj.size(); ++i) who[tin[i]] = i;
35        }
36
37        vector<pair<int, int>> path(int u, int v) {
38            vector<pair<int, int>> > res;
39            for (; v = par[head[v]]) {
40                if (depth[head[u]] > depth[head[v]]) swap(u, v);
41                if (head[u] != head[v]) {
42                    res.emplace_back(tin[head[v]], tin[v]);
43                } else {
44                    if (depth[u] > depth[v]) swap(u, v);
45                    res.emplace_back(tin[u], tin[v]);
46                    return res;
47                }
48            }
49
50            pair<int, int> subtree(int u) {
51                return {tin[u], tout[u]};
52            }
53
54            int dist(int u, int v) {
55                return depth[u] + depth[v] - 2 * depth[lca(u, v)];
56            }
57
58            int lca(int u, int v) {
59                for (; v = par[head[v]]) {
60                    if (depth[head[u]] > depth[head[v]]) swap(u, v);
61                    if (head[u] == head[v]) {
62                        if (depth[u] > depth[v]) swap(u, v);
63                        return u;
64                    }
65                }
66            }
67
68            bool isAncestor(int u, int v) {
69                return tin[u] <= tin[v] && tout[u] >= tout[v];
70            }
71        };

```

9.4 Centroid Decomposition

9.4.1 CentroidDecomposition

```

1 class CentroidDecomposition {
2 private:
3     int n;
4     vector<vector<int>> adj;
5     vector<int> sz, par, nodes;
6     vector<bool> vis;
7
8     int dfsSz(int u, int p) {
9         sz[u] = 1;
10        for (int v : adj[u]) {
11            if (vis[v] || v == p) continue;
12            sz[u] += dfsSz(v, u);
13        }
14        return sz[u];
15    }
16
17    int getCentroid(int u, int p, int total_nodes) {
18        for (int v : adj[u]) {
19            if (vis[v] || v == p) continue;
20            if (sz[v] * 2 > total_nodes) {
21                return getCentroid(v, u, total_nodes);
22            }
23        }
24        return u;
25    }
26
27    void collect(int u, int p) {
28        nodes.push_back(u);
29        for (int v : adj[u]) {
30            if (vis[v] || v == p) continue;
31            collect(v, u);
32        }
33    }
34
35    void build(int u, int p) {
36        int total_nodes = dfsSz(u, p);
37        int centroid = getCentroid(u, p, total_nodes);
38
39        if (p != -1) {
40            par[centroid] = p;
41        } else {
42            par[centroid] = centroid;
43        }
44
45        vis[centroid] = true;
46
47        // =====
48        // PROBLEM SPECIFIC PROCESSING
49        // =====
50        // Process paths going through the 'centroid' here
51        for (int v : adj[centroid]) {
52            if (vis[v]) continue;
53            nodes.clear();
54            collect(v, centroid);
55
56            // e.g., Update answers using the nodes collected from this subtree branch
57        }
58        // =====
59
60        // Recursively decompose the remaining components
61        for (int v : adj[centroid]) {
62            if (vis[v]) continue;

```



```

63         build(v, centroid);
64     }
65 }
66
67 public:
68     // Initialize with 1-based indexing
69     CentroidDecomposition(int nodes_count) : n(nodes_count) {
70         adj.assign(n + 1, vector<int>());
71         sz.assign(n + 1, 0);
72         par.assign(n + 1, 0);
73         vis.assign(n + 1, false);
74     }
75
76     void add_edge(int u, int v) {
77         adj[u].push_back(v);
78         adj[v].push_back(u);
79     }
80
81     // Start decomposition, usually from node 1
82     void decompose(int root = 1) {
83         build(root, -1);
84     }
85
86     // Returns the parent of a node in the centroid tree
87     int get_centroid_parent(int u) const {
88         return par[u];
89     }
90 };

```

10 Game Theory

11 Geometry

11.1 Geometry 2D

11.1.1 Point2D

```

1  template<class T>
2  int sgn(T x) { return (x > 0) - (x < 0); }
3
4  template<class T>
5  struct Point {
6      typedef Point P;
7      T x, y;
8
9      Point(T x = 0, T y = 0) : x(x), y(y) {}
10
11      bool operator<(P p) const { return tie(x, y) < tie(p.x, p.y); }
12
13      bool operator==(P p) const { return tie(x, y) == tie(p.x, p.y); }
14
15      P operator+(P p) const { return P(x + p.x, y + p.y); }
16
17      P operator-(P p) const { return P(x - p.x, y - p.y); }
18
19      P operator*(T d) const { return P(x * d, y * d); }
20
21      P operator/(T d) const { return P(x / d, y / d); }
22
23      T dot(P p) const { return x * p.x + y * p.y; }
24
25      T cross(P p) const { return x * p.y - y * p.x; }
26
27      T cross(P a, P b) const { return (a - *this).cross(b - *this); }
28
29      T dist2() const { return x * x + y * y; }
30
31      T dist2(P b) const { return (*this - b).dist2(); }
32
33      double dist() const { return sqrt((double) dist2()); }
34
35      double dist(P b) { return (*this - b).dist(); }
36
37      // angle to x-axis in interval [-pi, pi]
38      double angle() const { return atan2(y, x); }
39
40      P unit() const { return *this / dist(); } // makes dist()=1
41      P perp() const { return P(-y, x); } // rotates +90 degrees
42      P normal() const { return perp().unit(); }
43
44      P getVector(const P &p) const { return p - (*this); }
45
46      // returns point rotated 'a' radians ccw around the origin
47      P rotate(double a) const {
48         return P(x * cos(a) - y * sin(a), x * sin(a) + y * cos(a));
49     }
50
51     friend ostream &operator<<(ostream &os, P p) {
52         // return os << "(" << p.x << ", " << p.y << ")";
53         return os << p.x << " " << p.y;
54     }
55
56     friend istream &operator>>(istream &is, P &p) {
57         return is >> p.x >> p.y;
58     }
59
60
61     // Project point onto line through a and b (assuming a != b).
62     P projectOnLine(const P &a, const P &b) const {
63         P ab = a.getVector(b);
64         P ac = a.getVector(*this);
65         return a + ab * ac.dot(ab) / a.dist2(b);
66     }
67
68     // Project point c onto line segment through a and b (assuming a != b).
69     P projectOnSegment(const P &a, const P &b) const {
70         P &c = (P &) *this;
71         P ab = a.getVector(b);
72         P ac = a.getVector(c);
73
74         long double r = ac.dot(ab), d = a.dist2(b);
75         if (r < 0) return a;
76         if (r > d) return b;

```

```

77         return a + ab * r / d;
78     }
79
80     P reflectAroundLine(const P &a, const P &b) const {
81         return projectOnLine(a, b) * 2 - (*this);
82     }
83
84 };
85
86 int dcmp(const ld &a, const ld &b){
87     if(fabs(a - b) < eps)
88         return 0;
89
90     return (a > b ? 1 : -1);
91 }

```

11.1.2 Line2D

```

1  template<class T>
2  double lineDist(T p, T s, T e) {
3      if (s == e) {
4         return s.dist(p);
5     }
6     return fabs((p - s).cross(e - s) / (e - s).dist());
7 }
8
9  template<class T>
10 pair<int, T> lineInter(T s1, T e1, T s2, T e2) {
11     // first = 0 no intersection
12     // first = 1 intersection
13     // first = -1 infinite intersection
14     auto d = (e1 - s1).cross(e2 - s2);
15     if (dcmp(d, 0) == 0) // if parallel same line first = -1 else
16         first = 0
17         return {-(s1.cross(e1, s2) == 0), {0, 0}};
18     auto p = s2.cross(e1, e2), q = s2.cross(e2, s1);
19     return {1, (s1 * p + e1 * q) / d};
20
21 }
22
23 template<class T>
24 bool onSegment(T p, T s, T e) {
25     return dcmp(p.cross(s, e), 0) == 0 && dcmp((s - p).dot(e - p), 0)
26         <= 0;
27
28 }
29
30 template<class T>
31 double segDist(T p, T s, T e) {
32     if (dcmp((p - s).dot(e - s), 0) <= 0) return s.dist(p);
33     if (dcmp((p - e).dot(e - s), 0) >= 0) return e.dist(p);
34     return lineDist(p, s, e);
35 }
36
37 template<class T>
38 pair<int, T> segInter(T s1, T e1, T s2, T e2) {
39     // first = 0 no intersection
40     // first = 1 intersection
41     // first = -1 infinite intersection
42     pair<int, T> ret = lineInter(s1, e1, s2, e2);
43     if (ret.first == 0) return ret;
44     else if (ret.first == 1) {
45         if (onSegment(ret.second, s1, e1) && onSegment(ret.second, s2, e2))
46             return ret;
47         else
48             return {0, {0, 0}};
49     } else {
50         if (onSegment(s1, s2, e2) || onSegment(e1, s2, e2))
51             return {-1, (onSegment(s1, s2, e2) ? s1 : e1)};
52         else if (onSegment(s2, s1, e1) || onSegment(e2, s1, e1))
53             return {-1, (onSegment(s2, s1, e1) ? s2 : e2)};
54         else
55             return {0, {0, 0}};
56     }
57 }
58
59 template<class T>
60 T closestOnSegment(T p, T s, T e) {
61     if ((p - s).dot(e - s) <= 0) return s;
62     else if ((p - e).dot(e - s) >= 0) return e;
63     else return p.projectOnLine(s, e);
64 }
65
66 template<class T>
67 double segSegDist(T s1, T e1, T s2, T e2) {
68     if (segInter(s1, e1, s2, e2).first != 0)
69         return 0;
70     double ret = min({segDist(s1, s2, e2), segDist(e1, s2, e2), segDist(s2, s1, e1), segDist(e2, s1, e1)});
71     return ret;
72 }
73
74
75
76 template<class T>
77 bool onRay(T p, T s, T e) {
78     return dcmp(p.cross(s, e), 0) == 0 && dcmp((p - s).dot(e - s), 0)
79         >= 0;
80
81 }
82
83 template<class T>
84 double rayDist(T p, T s, T e) {
85     if ((p - s).dot(e - s) <= 0) {
86         return s.dist(p);
87     }
88     return lineDist(p, s, e);
89 }
90
91 template<class T>
92 pair<int, T> rayInter(T s1, T e1, T s2, T e2) {

```

```

92 // first = 0 no intersection
93 // first = 1 intersection
94 // first = -1 infinite intersection
95 pair<int, T> ret = lineInter(s1, e1, s2, e2);
96 if (ret.first == 0) return ret;
97 else if (ret.first == 1) {
98     if (onRay(ret.second, s1, e1) && onRay(ret.second, s2, e2))
99         return ret;
100     else
101         return {0, {0,0}};
102 } else {
103     if(onRay(s1, s2, e2) || onRay(s2, s1, e1))
104         return {-1,onRay(s1, s2, e2) ? s1:s2};
105     else
106         return {0, {0,0}};
107 }
108 }
109
110 template<class T>
111 double rayRayDist(T s1, T e1, T s2, T e2) {
112     if(rayInter(s1,e1,s2,e2).first != 0)
113         return 0;
114     double ret = min(rayDist(s1,s2,e2), rayDist(s2,s1,e1));
115     return ret;
116 }

```

11.1.3 AngleOperations

```

1 template<class T>
2 // angle between [0, 2*pi]
3 ld angleBetween(T a, T b) {
4     ld ret = atan2(a.cross(b), a.dot(b));
5     if (dcmp(ret, 0) == -1) {
6         ret += 2 * PI;
7     }
8     // return min(ret, 2 * PI - ret); //to return the smaller angle
9     return ret;
10 }
11
12 template<class T>
13 ld angle0(T a, T O, T b) { /// angle(aOb)
14     assert(a.dist(O) > eps && b.dist(O) > eps); // nan
15     T v1 = (a - O), v2 = (b - O);
16     return angleBetween(v1, v2);
17 }
18
19 // [-pi, pi]
20 ld format(ld angle) {
21     while (dcmp(angle, PI) == 1)
22         angle -= 2 * PI;
23     while (dcmp(angle, -PI) != 1)
24         angle += 2 * PI;
25     return angle;
26 }
27 }

```

11.1.4 PolygonArea

```

1 template<class T>
2 ld polygonArea(vector<T>&v){
3     ld ret = 0;
4     int n = v.size();
5     for (int i = 0; i < n; ++i) {
6         ret += v[i].cross(v[(i+1)%n]);
7     }
8     return 0.5 * abs(ret);
9 }

```

11.1.5 Circle3Points

```

1 typedef Point<double> P;
2
3 bool isColliner(const P &A, const P &B, const P &C)
4 {
5     return dcmp(P(B - A).cross(P(C - A)), 0) == 0;
6 }
7
8 P ccCenter(const P &A, const P &B, const P &C)
9 {
10     P b = C - A, c = B - A;
11     return A + (b * c.dist2() - c * b.dist2()).perp() / b.cross(c) / 2;
12 }
13
14 double ccRadius(const P &A, const P &B, const P &C)
15 {
16     return (B - A).dist() * (C - B).dist() * (A - C).dist() /
17         abs((B - A).cross(C - A)) / 2;
18 }

```

11.1.6 CircleLineIntersection

```

1 template<class T>
2 vector<T> circleLine(T o, double r, T a, T b) {
3     double h2 = r * r - lineDist(o, a, b) * lineDist(o, a, b);
4     if (dcmp(h2, 0) != -1) { // the line touches the circle
5         T p = o.projectOnLine(a, b); // point P
6         T h = (a - b).unit() * sqrt(h2); // vector parallel to l, of
7             // length h
8         if (p - h == p + h)
9             return {p - h};
10         else
11             return {p - h, p + h};
12     }
13     return {};
14 }
15
16 template<class T>
17 vector<T> circleSegment(T o, double r, T a, T b) {
18     vector<T> v = circleLine(o, r, a, b);
19     vector<T> out;
20     for (auto x: v)
21         if (onSegment(x, a, b))
22             out.pb(x);

```

```

22     return out;
23 }

```

11.1.7 CircleCircleIntersection

```

1 template<class T>
2 ld circleInter(T a,T b,ll r1,ll r2){
3     if(r1 > r2){
4         swap(r1,r2);
5         swap(a,b);
6     }
7     ld d = sqrt((a.x - b.x) * (a.x - b.x) + (a.y - b.y) * (a.y - b.y));
8     if(d >= r1 + r2){
9         return 0;
10    }
11    if(d + r1 <= r2){
12        return PI * r1 * r1;
13    }
14    ld theta = acos((d * d + r2*r2 - r1*r1)/(2 * d * r2));
15    ld alpha = acos((d * d + r1*r1 - r2*r2)/(2 * d * r1));
16    ld s1 = theta * r2 * r2;
17    ld t1 = 0.5 * r2 * r2 * sin(2 * theta);
18    ld s2 = alpha * r1 * r1;
19    ld t2 = 0.5 * r1 * r1 * sin(2 * alpha);
20    return (s1 - t1) + (s2 - t2);
21 }

```

11.1.8 ConvexHullAndrew

```

1 template<class T>
2 vector<T> convexHull(vector<T> pts, bool inc_collinear = false) {
3
4     T p0 = *min_element(pts.begin(), pts.end(), [](T &a, T &b) {
5         return make_pair(a.y, a.x) < make_pair(b.y, b.x);
6     });
7
8     sort(pts.begin(), pts.end(), [&p0](T &a, T &b) {
9         ll o = p0.cross(a, b);
10        if (o != 0) return o > 0;
11        return (a - p0).dist2() < (b - p0).dist2();
12    });
13
14    if (inc_collinear) {
15        int ind = pts.size() - 1;
16        while (ind >= 0 && p0.cross(pts[ind], pts.back()) == 0) --ind;
17        reverse(pts.begin() + ind + 1, pts.end());
18    }
19
20    vector<T> ch;
21    for (int i = 0; i < (int) pts.size(); i++) {
22        int sz = ch.size();
23        while (ch.size() > 1 &&
24            (ch[sz - 2].cross(ch[sz - 1], pts[i]) < 0 ||
25             (!inc_collinear && ch[sz - 2].cross(ch[sz - 1], pts[i])
26              == 0))) {
27            ch.pop_back();
28            sz = ch.size();
29        }
30        ch.push_back(pts[i]);
31    }
32    return ch;
33 }

```

11.1.9 RotatingCalipers

```

1 template<class T>
2 ll hullDiameter(vector<T> S) {
3     int n = S.size(), j = n < 2 ? 0 : 1;
4     ll ret = 0;
5     for (int i = 0; i < j; ++i) {
6         for (; j = (j + 1) % n) {
7             ret = max(ret, (ll)(S[i] - S[j]).dist2());
8             if ((S[(j + 1) % n] - S[j]).cross(S[i + 1] - S[i]) >= 0)
9                 break;
10        }
11    }
12    // returns the squared diameter
13    return ret;
14 }
15
16 template<class T>
17 ld hullWidth(vector<T> S) {
18     int n = S.size();
19     if(n <= 2) return 0;
20     int i = 0, j = 1;
21     ld ret = 1e18;
22     while (i < n) {
23         while((S[(i + 1) % n] - S[i]).cross(S[(j + 1)%n] - S[j]) >= 0) j
24             = (j + 1) % n;
25         ret = min(ret, lineDist(S[j], S[i], S[(i+1)%n]));
26         i++;
27     }
28     return ret;

```

11.1.10 MinimumEnclosingCircle

```

1 typedef Point<double> P;
2 #define rep(aa, bb, cc) for(int aa = bb; aa < cc; aa++)
3
4 P ccCenter(const P &A, const P &B, const P &C) {
5     P b = C - A, c = B - A;
6     return A + (b * c.dist2() - c * b.dist2()).perp() / b.cross(c) / 2;
7 }
8
9 pair<P, double> mec(vector<P> ps) {
10     shuffle(all(ps), mt19937(time(0)));
11     P o = ps[0];
12     double r = 0, EPS = 1 + 1e-8;
13     rep(i, 0, ps.size()) if ((o - ps[i]).dist() > r * EPS) {
14         o = ps[i], r = 0;
15         rep(j, 0, o) if ((o - ps[j]).dist() > r * EPS) {

```

```

16         o = (ps[i] + ps[j]) / 2;
17         r = (o - ps[i]).dist();
18         rep(k, 0, j) if ((o - ps[k]).dist() > r * EPS) {
19             o = ccCenter(ps[i], ps[j], ps[k]);
20             r = (o - ps[i]).dist();
21         }
22     }
23 }
24 return {o, r};
25 }

```

11.1.11 HalfPlaneIntersection

```

1 // Basic half-plane struct.
2 template<class T>
3 struct Halfplane {
4
5     // 'p' is a passing point of the line and 'pq' is the direction
6     // vector of the line.
7     T p, pq;
8     long double angle;
9
10     Halfplane() {}
11     Halfplane(const T& a, const T& b) : p(a), pq(b - a) {
12         angle = atan2l(pq.y, pq.x);
13     }
14
15     // Check if point 'r' is outside this half-plane.
16     // Every half-plane allows the region to the LEFT of its line.
17     bool out(const T& r) {
18         return dcmp(pq.cross(r - p), 0) < 0;
19     }
20
21     // Comparator for sorting.
22     bool operator < (const Halfplane& e) const {
23         return angle < e.angle;
24     }
25
26     // Intersection point of the lines of two half-planes. It is
27     // assumed they're never parallel.
28     friend T inter(const Halfplane& s, const Halfplane& t) {
29         long double alpha = (ld)(t.p - s.p).cross(t.pq) / s.pq.cross(t.pq);
30         return s.p + (s.pq * alpha);
31     }
32 };
33
34 template<class T>
35 vector<T> hpIntersect(vector<Halfplane<T>> H) {
36     ld inf = 1e9;
37     T box[4] = { // Bounding box in CCW order
38         Point(inf, inf),
39         Point(-inf, inf),
40         Point(-inf, -inf),
41         Point(inf, -inf)
42     };
43
44     for(int i = 0; i < 4; i++) { // Add bounding box half-planes.
45         Halfplane aux(box[i], box[(i+1) % 4]);
46         H.push_back(aux);
47     }
48
49     // Sort by angle and start algorithm
50     sort(H.begin(), H.end());
51     deque<Halfplane<T>> dq;
52     int len = 0;
53     for(int i = 0; i < int(H.size()); i++) {
54
55         // Remove from the back of the deque while last half-plane is
56         // redundant
57         while (len > 1 && H[i].out(inter(dq[len-1], dq[len-2]))) {
58             dq.pop_back();
59             --len;
60         }
61
62         // Remove from the front of the deque while first half-plane is
63         // redundant
64         while (len > 1 && H[i].out(inter(dq[0], dq[1]))) {
65             dq.pop_front();
66             --len;
67         }
68
69         // Special case check: Parallel half-planes
70         if (len > 0 && fabs1(H[i].pq.cross(dq[len-1].pq)) < eps) {
71             // Opposite parallel half-planes that ended up checked
72             // against each other.
73             if (dcmp(H[i].pq.dot(dq[len-1].pq), 0) < 0)
74                 return vector<T>();
75
76             // Same direction half-plane: keep only the leftmost half-
77             // plane.
78             if (H[i].out(dq[len-1].p)) {
79                 dq.pop_back();
80                 --len;
81             }
82             else continue;
83         }
84
85         // Add new half-plane
86         dq.push_back(H[i]);
87         ++len;
88     }
89
90     // Final cleanup: Check half-planes at the front against the back
91     // and vice-versa
92     while (len > 2 && dq[0].out(inter(dq[len-1], dq[len-2]))) {
93         dq.pop_back();
94         --len;
95     }
96
97     while (len > 2 && dq[len-1].out(inter(dq[0], dq[1]))) {
98         dq.pop_front();
99         --len;
100     }
101 }

```

```

94 // Report empty intersection if necessary
95 if (len < 3) return vector<T>();
96
97 // Reconstruct the convex polygon from the remaining half-planes.
98 vector<T> ret(len);
99 for(int i = 0; i < len; i++) {
100     ret[i] = inter(dq[i], dq[(i+1)%len]);
101 }
102 return ret;
103 }
104 }

```

11.2 Geometry 3D

11.2.1 Point3D

```

1 template<class T>
2 struct Point3D {
3     typedef Point3D P;
4     typedef const P &R;
5     T x, y, z;
6
7     explicit Point3D(T x = 0, T y = 0, T z = 0) : x(x), y(y), z(z) {}
8
9     bool operator<(R p) const {
10         return tie(x, y, z) < tie(p.x, p.y, p.z);
11     }
12
13     bool operator==(R p) const {
14         return tie(x, y, z) == tie(p.x, p.y, p.z);
15     }
16
17     P operator+(R p) const { return P(x + p.x, y + p.y, z + p.z); }
18
19     P operator-(R p) const { return P(-x, -y, -z); }
20
21     P operator-(R p) const { return P(x - p.x, y - p.y, z - p.z); }
22
23     P operator*(T d) const { return P(x * d, y * d, z * d); }
24
25     P operator/(T d) const { return P(x / d, y / d, z / d); }
26
27     T dot(R p) const { return x * p.x + y * p.y + z * p.z; }
28
29     P cross(R p) const {
30         return P(y * p.z - z * p.y, z * p.x - x * p.z, x * p.y - y * p.x);
31     }
32
33     T dist2() const { return x * x + y * y + z * z; }
34
35     double dist() const { return sqrt((double) dist2()); }
36
37     //Azimuthal angle ( longitude ) to x=axis in interval [pi , pi ]
38     double phi() const { return atan2(y, x); }
39
40     //Zenith angle ( latitude ) to the z=axis in interval [0 , pi ]
41     double theta() const { return atan2(sqrt(x * x + y * y), z); }
42
43     P unit() const { return *this / (T) dist(); } //makes dist ()=1
44     //returns unit vector normal to *this and p
45     P normal(P p) const { return cross(p).unit(); }
46
47     //returns point rotated 'angle ' radians ccw around axis
48     P rotate(double angle, P axis) const {
49         double s = sin(angle), c = cos(angle);
50         P u = axis.unit();
51         return u * dot(u) * (1 - c) + (*this) * c - cross(u) * s;
52     }
53
54     friend ostream &operator<<(ostream &os, P p) {
55         // return os << "(" << p.x << ", " << p.y << ")";
56         return os << p.x << " " << p.y << " " << p.z;
57     }
58
59     friend istream &operator>>(istream &is, P &p) {
60         return is >> p.x >> p.y >> p.z;
61     }
62 };
63
64 namespace Axis {
65     Point3D<double> x(1, 0, 0);
66     Point3D<double> y(0, 1, 0);
67     Point3D<double> z(0, 0, 1);
68 }
69

```